Master Thesis

Preserving Privacy in XAI: An Exploration of the Interplay between Differential Privacy and Federated Learning

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Abstract

The digitization of health data offers significant opportunities for enhancing medical diagnostics and research, but it also raises concerns about data privacy and security, particularly with the use of machine learning models. This thesis addresses the critical challenge of balancing model explainability with data privacy by investigating the effects of Differential Privacy (DP) and Federated Learning (FL) on machine learning models.

The study employs the MIMIC dataset to train logistic regression and neural network models, assessing various configurations of DP privacy budgets and FL client numbers. It evaluates the impact of these privacy-preserving techniques on model performance, explainability, and resilience to Membership Inference Attacks (MIA). The findings reveal that DP's impact on performance is model-dependent, with neural networks being more affected than logistic models. FL generally maintains performance, but has minimal effects beyond a certain client threshold. Explainability metrics remain largely unaffected by DP and FL, though combining these techniques can mitigate some of DP's negative effects. Privacy evaluations show that while DP does not significantly alter MIA success rates, FL reduces attack precision, thus enhancing privacy.

This research suggests that there is no clear trade-off between privacy and explainability, as the effects are highly model-dependent. These insights still provide actionable guidance for deploying privacy-preserving machine learning models, ensuring both interpretability and robust data protection.

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Chapter 1

Introduction

The digitisation of health data has initiated a new era of medical research and healthcare delivery, offering unprecedented opportunities for generating new knowledge, improved diagnostics by propagating new discoveries, streamlined healthcare and advancing research processes [1, 2]. As health data becomes increasingly digitised, it can be leveraged to develop sophisticated machine learning models that can classify diseases and enhance diagnostic accuracy for multiple medical problems [3]. However, the extensive use of this data also raises significant concerns about privacy and security, particularly in light of recent findings that neural networks are prone to unintended memorization and data leakage. Neural networks can inadvertently memorize and leak sensitive information such as personal text messages and credit card numbers and reveal it during inference, e.g. through text completion functionalities [4]. This revelation underscores the critical need for robust techniques that preserve privacy of individuals in the development and deployment of machine learning models, especially in the healthcare domain where data sensitivity is paramount.

In response to these privacy concerns, regulatory frameworks such as the General Data Protection Regulation (GDPR) [5] and the AI Act [6] in the European Union and the Health Insurance Portability and Accountability Act (HIPAA) [7] in the United States have been established to protect personal data. The GDPR and AI Act, in particular, impose stringent requirements on organizations that process personal data, including provisions that could mandate AI providers to offer explanations for automated decisions affecting individuals. These regulations emphasize the necessity of ensuring both data privacy and model explainability in AI systems, posing a significant challenge for researchers and practitioners.

To address these challenges, privacy-preserving techniques such as Differential Privacy and Federated Learning have emerged as promising solutions. Differential Privacy introduces noise into the data to obscure individual contributions, therefore providing strong privacy guarantees while allowing meaningful insights to be extracted from the data. Federated Learning, on the other hand, enables multiple parties to collaboratively train a machine learning model without sharing their raw data, thus maintaining data locality and enhancing privacy.

The enhancement of privacy however has its drawbacks. The literature hints that applying Differential Privacy or Federated Learning can harm the explainability of a model, and therefore establishing a trade-off between the privacy and explainability of a model [8, 9]. Lowering the explainability of a model can be detrimental for the application of this model, e.g. in the medical domain where the patient has to be informed about their procedures and why they are undertaken. If the explainability of a model is reduced in that setting, the patient can lose confidence in a treatment he might desperately need. Hence, it can be challenging to apply these

privacy-preserving techniques and they have to be tailored to the domain and problem they are applied to.

Despite the advancements in Differential Privacy and Federated Learning, these techniques are not immune to adversarial attacks. Adversarial attacks are utilised to extract information of the target model about either the data it was trained on or the model architecture itself. Among the various attack methods, Membership Inference Attacks (MIAs) are deserving particular attention, due to their capability of extracting information from the original dataset. MIAs aim to determine whether a specific data point was included in the training set of a machine learning model, potentially exposing sensitive information and violating privacy guarantees.

Differential Privacy and Federated Learning have both separately and in combination been investigated on their contribution to privatising data [9, 10]. Furthermore, the effect of these techniques on the quality of explanations have been explored [11] and how explanations in general can be exploited for adversarial attacks [12]. However, to the best of my knowledge, it has never been empirically quantified how Differential Privacy and Federated Learning can maintain explainability while also minimising the success rate of an actual attack. This poses a gap in research addressed by this work. Therefore, this work explores the intricate balance between maintaining model explainability and ensuring data privacy in the context of the privacy-preserving techniques Differential Privacy and Federated Learning, with a specific focus on exploring the risks posed by MIAs. By exploring the trade-offs and synergies between these objectives, this research aims to provide clear recommendations for developing AI systems that are both interpretable and secure, ultimately contributing to the responsible and ethical use of health data in machine learning applications.

1.1 Research questions

To the best of my knowledge, there is no empirical study investigating both the explainability and privacy through an empirical study using an actual MIA to qualify the privatisation abilities of Federated Learning and Differential Privacy. The literature suggests that there is a trade-off between preserving explainability and performance while privatising a model. However, none of these studies investigated this privacy-explainability trade-off with the application of Differential Privacy and Federated Learning and then being attacked by a MIA. Therefore, in order to close this gap, the following questions are proposed and answered in this work:

1. What is the effect of applying Differential Privacy (DP) and/or Federated Learning (FL) on the explainability of a classification model?
2. To what extent can the explainability of models trained with DP and/or FL be preserved, while concurrently minimising the success rate of external attackers attempting to infer information about the training data through a black-box membership inference attack?

The goal of RQ 1 is to provide insights to how privacy-preserving techniques like Differential Privacy and Federated Learning affect the explainability of a machine learning model. Therefore, addressing the explainability part of the privacy-explainability trade-off. To analyse the privacy part of this trade-off. RQ 2 aims at exploring to what extent explainability can be maintained when data privacy is challenged by an adversarial attack.

1.2 Outline

This thesis is structured as follows: In Chapter 2, an overview on the current state of research related to this work is given. Additionally, all used methods are listed and described. Afterwards,

Chapter 3 outlines how this work aims to answer the research questions posted in Section 1.1 by conducting empirical experiments. The results of these experiments are described in Chapter 4 and discussed in Chapter 5. Finally, Chapter 6 closes by drawing conclusions from the results discussed and providing recommendations for future work.

Chapter 2

Background and related work

This chapter first introduces methods and techniques used in this work. With this background established, related literature is reviewed to contextualise this work.

2.0.1 Performance Metrics

With the primary focus of this thesis being to evaluate the explainability and susceptibility to attacks of the classifiers, their performance is still providing important context for the other findings. To measure performance, Accuracy, F-Score and AUROC were considered, as they are widely used in the literature and are utilised in this work to compare the training results to the mentioned benchmarks in Section 3.3.1. All of these metrics are based on the confusion matrix in the binary classification case, as shown in Table 2.1. Here, the prefix true and false determine whether the prediction of the classifier was correct or not. Hence, the samples declared as true positives were correctly labelled with a positive label. Samples that are classified as positive, but are actually negative, are called false positives and vice versa for the samples labelled as negative by the predictor.

		Predicted labels	
		positive	negative
True labels	positive	true positive (TP)	false negative (FN)
	negative	false positive (FN)	true negative (TN)

Table 2.1: Confusion matrix for binary classification

Accuracy

The accuracy of a model describes how well the classifier could predict the correct label for all samples. Accuracy is defined as in Eq. 2.1

$$Accuracy = \frac{TP + TN}{\#samples} \tag{2.1}$$

F-Score

The F-Score [13] (Eq. 2.4) combines Precision (Eq. 2.2) and Recall (Eq. 2.3). Precision qualifies how many positive labels are correctly given by the classifier. Whereas, recall measures how well the classifier found all samples that should be labelled positive.

$$Precision = \frac{TP}{TP + FP} \tag{2.2}$$

$$Recall = \frac{TP}{TP + FN} \tag{2.3}$$

With combining Precision and Recall, the F-Score gives a good indication on how accurate the positive predictions of a classifier are.

$$F\text{-}Score = \frac{(1 + \beta^2) \cdot Precision \cdot Recall}{(\beta^2 \cdot Precision) + Recall} = \frac{(\beta^2 + 1) \cdot TP}{(\beta^2 + 1) \cdot TP + \beta^2 FN + FP} \tag{2.4}$$

where β is a parameter giving importance to either Precision or Recall, depending on its value. During the evaluation of the classifiers, the F-Score with $\beta = 1$ is used, which gives equal importance to Precision and Recall. This is also called the F1-Score.

Area under the ROC curve

The receiver operating characteristic (ROC) [14] is a standard metric used in the medical sciences [13]. It is used to qualify the ability of preventing false classification of a model. Formally, the ROC curve is build by creating pairs of Recall (see Eq. 2.3) and the false positive rate (FPR) (Eq. 2.5) for different decision thresholds. This process can be formally defined by Def. 1 and can be displayed with Recall and FPR as its dimensions.

$$FPR = \frac{FP}{TN + FP} \tag{2.5}$$

Definition 1 *Let $\{\theta_1, \theta_2, ..., \theta_n\}$ be the set of prediction thresholds that ranges over all unique prediction scores of a classifier C and $Recall_{\theta_i}, FPR_{\theta_i}$ be the Recall and FPR of C with prediction threshold θ_i, then the set of pairs $\{(Recall_{\theta_i}, FPR_{\theta_i}) \mid i \in \{1, ..., n\}\}$ describes a ROC curve for C.*

The area under the ROC curve (AUROC) (Eq. 2.6) summarises the results from the ROC metric by calculating the area under the ROC curve by applying the Trapezoidal rule. This approximates the area by summing up the areas of trapezoids formed between consecutive points on the ROC curve. This can be described succinctly in a continuous fashion, as in Eq. 2.6. [13]

$$AUROC = \int_0^1 Recall \; dFPR \tag{2.6}$$

With this aggregation, AUROC represents the ROC curve as a single number for easier comparison and quantifies how well the classifier can avoid false classification.

2.0.2 Paired t significance test

In order to test whether the results of two different models are statistically significant, the paired t significance test is utilised as described in [15]. Therefore, two observations of two different models are investigated, with the null hypothesis that the mean between the paired observations is 0, i.e. $\mu_0 = 0$. First, the t-statistic is calculated as shown in Eq. 2.7, where $\bar{x}$ denotes the mean of the differences of the paired observations, s is the standard deviation of these differences and n is the number of samples.

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \tag{2.7}$$

Since, the goal of this evaluation is to test whether the means of the two observations are different, a two-tailed test is employed. Hence, the p-value is calculated as defined in Eq. 2.8, where T follows a t-distribution with $n - 1$ degrees of freedom.

$$p = 2 \cdot P(T \geq |t| \mid \mu_0 = 0) \tag{2.8}$$

This test therefore describes a probability of whether the null hypothesis of the mean between the paired observations is zero, i.e. if the difference of the two observations is statistically significant or not. For this evaluation, a threshold of $p < 0.05$ for statistical significance is used, as is common in the literature.

2.1 Privacy-preserving techniques

Protecting the identity of a person has become important. Since, it is possible to infer data though a published machine learning model [16], it has become a challenge to protect these identities. Especially in medical studies or the collection of data for datasets in the health domain, the usage of the personal data should not have an impact on the patients' privacy. If it is possible for an attacker to extract personal information from a publicly available dataset, the right of protection of personal data is no longer given. Furthermore, the patient might be affected by it as they are not the only one governing over their personal data. Furthermore, attributes of a person can even be extracted if the dataset does not contain data to reidentify this person, e.g. by connecting this data to different datasets. Hence, several regulations have been set in place by different parts of the world.

In the European Union, the General Data Protection Regulation (GDPR) [5] is applicable as of 2018 and "protects fundamental rights and freedoms of [...] persons and in particular their right to the protection of personal data" (Art. 1, §2). Additionally, it enforces "Privacy by Design", which means privacy should already be integrated in the design of technology. Therefore, technologies that protect the privacy of personal data should be fundamental for every technical application. Furthermore, it classifies medical data to be protected by prohibiting the processing of data for uniquely identifying a person and data concerning health (Art. 9, §1).

This global view on the protection of personal data only gets strengthened by the Health Insurance Portability and Accountability Act (HIPAA) [7] in the United States, enacted in 1996. With this law, "individually identifiable health information" is to be protected. This additionally focuses on health data specifically to further secure the privacy of personal data.

These regulations establish a need for techniques capable of safeguarding personal data, a requirement that extends to machine learning models. According to the fundamental law of recovery, providing overly accurate answers to too many questions will inevitably compromise privacy dramatically [17]. Consequently, algorithmic research on privacy should focus on delaying

this inevitability for as long as possible. Privacy-preserving techniques have thus been developed to support this goal by protecting personal data in combination with deidentifying practices, e.g. anonymisation and pseudonymisation. Differential Privacy and Federated Learning are two techniques that will be evaluated in this work. Their technical details are described in the following sections.

2.1.1 Differential Privacy

The goal of research on Differential Privacy is to delay the inevitability of compromising privacy of data as long as possible. Consequently, the same conclusions should be reached without regard to individuals taking part in collecting their data or not. Hence, the absence or altering of one sample in a dataset should not affect the decision outcome of a trained machine learning model and therefore protecting the privacy of a person represented by that sample. This is ensured by defining a parameter ϵ which is a measure for how private and less accurate predictions of a model are. This parameter lays down the foundation for the definition of Differential Privacy [18][19][20].

Definition 2 *Let D be the set of all Datasets. A randomised function $f : D \to R^d$ gives ϵ-Differential Privacy if for all datasets $D_1, D_2 \in D$ differing in at most one element and $\mathcal{S} \subseteq R^d$,*

$$\Pr[f(D_1) \in \mathcal{S}] \leq \exp(\epsilon) \Pr[f(D_2) \in \mathcal{S}]$$
$$\ln \frac{\Pr[f(D_1) \in \mathcal{S}]}{\Pr[f(D_2) \in \mathcal{S}]} \leq \epsilon$$

Differential Privacy is therefore not an algorithm, but a definition. It describes a promise by the data holder to the data subject that a singular subject will not be affected by giving their personal data for e.g. a study.

More over, this now begs the question on how to achieve ϵ-Differential Privacy for this function f. Dwork proves in her paper Differential Privacy that this is achieved by adding random noise to the prediction a made by f. The magnitude of this noise is chosen as a function of the Sensitivity of the function. Sensitivity is defined as follows [19].

Definition 3 *For $f : D \leftarrow R^d$ and for all datasets $D_1, D_2 \in D$ differing in at most one element, the L1-Sensitivity of f is*

$$\Delta f = \max_{D_1, D_2} \|f(D_1) - f(D_2)\|_1$$

Hence, the Sensitivity of the function f is described by the maximum change a single sample in the datasets could have on the output of the function f. It should be noted that this property of the function f is independent of the datasets used.

Adding noise to each component of the dataset with a scaled symmetric exponential distribution (e.g. Laplace or Gaussian) with variance σ^2 achieves $(\Delta f/\sigma)$-Differential Privacy, as proven by Dwork [19]. This privacy mechanism, denoted $\mathcal{K}_f$ is described by the following density function.

$$Pr[\mathcal{K}_f(X) = a] \propto \exp\left(-\frac{\|f(X) - a\|_1}{\sigma}\right)$$

Consequently, one must choose $\sigma \geq \epsilon/\Delta f$ to achieve ϵ-Differential Privacy. Additionally, it is worth noting that adding noise works best, when the Sensitivity Δf is small, because the change through adding noise has a lesser effect on the outcome of function f.

However, this strict definition of ϵ-Differential Privacy can be relaxed by adding a $\delta > 0$. With this relaxation, the "pure" ϵ-Differential Privacy can be broken with probability δ, which can be useful in unlikely cases, e.g. if the dataset is published in the clear. This changes the original definition as follows.[20][17][21]

Definition 4 *Let D be the set of all Datasets. A randomised function $f : D \to R^d$ gives (ϵ, δ)-Differential Privacy if for all datasets $D_1, D_2 \in D$ differing in at most one element and $\mathcal{S} \subseteq R^d$,*

$$\Pr[f(D_1) \in \mathcal{S}] \leq \exp(\epsilon) \Pr[f(D_2) \in \mathcal{S}] + \delta$$

This relaxation of the original definition is used e.g. in the application of Differential Privacy to machine learning [22], which is described below.

Differentially Private Stochastic Gradient Descent (DP-SGD)

In order to apply Differential Privacy to the machine learning field to fulfil the promise of data privacy, noise has to be added during one of the stages of the machine learning pipeline. Hence, it is only natural to investigate and change how the weights are updated when training a machine learning model. A common approach was introduced by Abadi et al. [22]. They developed an algorithm which changes stochastic gradient descent (SGD) by computing per-sample gradients clipping their L_2 norm, aggregating them and adding noise before updating the gradients. The exact from their paper is outlined in Algorithm 1.

Algorithm 1 Differentially private SGD

Input: Examples $X = \{x_1, ..., x_n\}$ with cardinality N, set of training steps $\mathcal{T} = \{0, ..., T\}$, weights θ, loss function $\mathcal{L}(\theta, x_i)$, where $\mathcal{L}(\theta, x_i)$ is the loss value for sample x_i. Parameters: learning rate η_t with $t \in \mathcal{T}$, noise scale σ, group size L, gradient norm bound C.
Initialise θ_0 randomly
for $t \in \mathcal{T}$ **do**
  Take a random sample $X_t \subseteq X$ with sampling probability L/N
  Compute gradient
  For each $x_i \in X_t$, compute $\mathbf{g}_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$
  Clip gradient
  $\bar{\mathbf{g}}_t(x_i) \leftarrow \mathbf{g}_t(x_i) / \max(1, \frac{\|\mathbf{g}_t(x_i)\|_2}{C})$
  Add noise
  $\tilde{\mathbf{g}}_t \leftarrow \frac{1}{L} \sum_i (\bar{\mathbf{g}}_t(x_i) + n_i)$, with $n_i \sim \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$
  Descent
  $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{\mathbf{g}}_t$
end for
Output θ_t and compute the overall privacy cost (ϵ, δ) using a privacy accounting method

For each step, first a gradient vector $\mathbf{g}_t$ is computed as in the original gradient descent algorithm. This vector then is clipped to prove the Differential Privacy guarantee, since there is no *a priori* bound on the size of the gradients. Adding noise to the clipped gradient vector is then a vital step for privatising the data before the new weights are assigned.

In this work, the python library Opacus [23] is used in order to utilise DP-SGD in the experiments.

With DP-SGD, machine learning algorithms can be trained in accordance to the requirements of Differential Privacy. Therefore, it is used in this work to account for different privacy demands that are further described in Section 3.3.

2.1.2 Federated Learning

The other privacy-preserving technique, that is investigated in this work, is Federated Learning introduced by McMahan et al. [24]. The origin of the idea comes from data already being naturally distributed, as most people have smartphones with personal data on it that they might not want to share without the promise of keeping their data private. Therefore, their idea is to leave the data where it is stored (e.g. on the mobile devices), training a local model on a so-called client and only aggregating the local models of a lot of different devices on a central server. Then, the data does not have to be transferred to the server to train a model, which can cause privacy issues. Rather, the model is trained locally and the updates are sent to the server to form an aggregated model. This server model then is sent back to the clients and the process starts again. This significantly reduces the attack surface to only the clients, rather than the devices and the server. Hence, Federated Learning is reducing the security risk by decoupling the model training from the need for direct access to the training data.

In their paper, they outline key properties of Federated Learning. First, it is **non-IID**[1], since the dataset of each client is tailored to this specific client and is therefore not representative of the entire client population distribution. This also play into the next property of the data being **unbalanced**, again leading to various amounts of data in the local datasets. They are also counting on the data being **massively distributed**, meaning that the average number of samples of each client are expected to be much smaller than the clients contributing to the server model. Finally, there is **limited communication** to be expected with then clients as they could be frequently offline, slow or the connection could be expensive.

Nonetheless, to establish an experiment environment, they fixed the number of clients participating in the learning process. Each of these clients possess their own local dataset. At the beginning of each training round, a random fraction of clients is selected which receive the current global model to train a local model on their individual data. The fraction of clients is selected here due to efficiency, because they found that training with all clients at each step does not automatically yield a large increase in model prediction performance. After the local model is trained on each client, the state of that model is sent as an update to the server. There, the updates coming from the clients are applied to the global model. This process is either repeated for a certain number of steps or until a learning goal is reached. This learning approach can be made more specific by defining how the global model is update. If the weights of the local models are aggregated with a weighted average, the algorithm is called FederatedAveraging (FedAvg). FedAvg is defined by the following parameters:

- A fraction C of clients is selected to train with gradient descent
- There are E steps of gradient descent performed by each client
- B denotes the local batch size

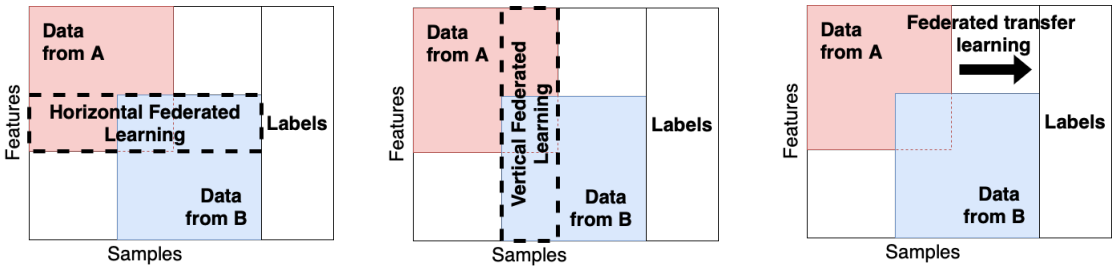
Having now described the fundamentals of the Federated Learning approach, there are different variants or attributes. Yang et al. [25] have described how privacy relates to Federated Learning and how different Federated Learning approach can be categorised. Their approach is described in the following section.

Distribution characteristics of Federated Learning

Federated Learning can be categorised in three different approaches to the algorithm: horizontal, vertical Federated Learning or federated transfer learning. These categories are formed based

[1] Independent and identically distributed

on the distribution characteristics of the data. All of these are described in the following and visualised in Figure 2.1.



(a) Horizontal FL (b) Vertical FL (c) Federated Transfer Learning

Figure 2.1: Categories of Federated Learning from [25]

If the data of the clients contain different sample identifiers, but share their feature space, it is called **horizontal or sample-based Federated Learning**, as visualised in Figure 2.1 (a). Applied to the real world, this could be two businesses in the same domain, but not being based at the same location. It could be a supermarket chain that has multiple stores at different locations of the country. Each store could represent a client. Since all of these stores are from the same supermarket chain, they have a largely overlapping feature space. However, due to the stores being at different locations, they do not share many of the same data subjects. In this scenario, it is assumed that only the server could violate data privacy, as it is typically assumed that the clients participating are honest. Therefore, one has to focus on security measures against an honest-but-curious server. At the end of training, all clients receive the full final server model with parameters.

With **vertical or feature-based Federated Learning**, as depicted in Figure 2.1 (b), the data from different clients used for training have the same sample identifiers or subjects. The feature space of these clients can be different. One example for this is: Two businesses are based in the same city, but do not share their domain, e.g. a bank and a supermarket. It is very likely that they have a big overlap in data subjects, since they are based in the same city, but do not share the same features since they come from different domains. Therefore, they could collaborate to achieve a common goal. With this scenario, honest-but-curious clients are assumed. An adversary could try to learn data from other clients by what is revealed by the inputs and outputs from the common model. As a countermeasure, sometimes a new semi-honest third party (STP) is introduced which does not share any common interests with the clients. This third party would then facilitate the computations between all clients to protect the data of each client. Secure Multiparty Computing, which is introduced later, provides proof of privacy for this communication. After the training has finished, each client holds the model parameters related to their own features. Hence, when the model is used for inference, all clients have to cooperate to generate output.

In contrast to horizontal and vertical Federated Learning, in **Federated Transfer Learning** the data of the clients not only differ in their sample identifiers, but also in their respective feature space. This results in a small intersection between data subject and features. Since this intersection is so small, a common representation between the different feature spaces of the participating clients is learned. This is achieved by using a limited set of common samples that are available to all clients. Through these common samples, the model can learn to map different

feature spaces into a shared representation. Once the common representation is established, the learned knowledge is transferred to make predictions for samples that have features from only one client. Hence, the model can still utilise the shared representation learned from the common samples. Typically, this method of Federated Learning involves two parties with a protocol similar to vertical Federated Learning. This security definition can then be extended.

Data privacy can not only be achieved by protocols or the different archetypes of Federated Learning, but additionally with several methods that are described next.

Privacy of Federated Learning

Differential Privacy as described in Section 2.1.1 can be combined with Federated Learning. Here, different techniques can be used to guarantee a certain degree of Differential Privacy. It can be used in the training process to protect data from each client by trying to obscure their input on the server model [10].

An extension to Federated Learning is to use **Homomorphic Encryption** [26]. During training, the parameters of the model that is trained are encrypted for the parameter exchange. Using encryption further improves data privacy during data exchange between clients and the server. Through this mechanism, other parties involved in the training process, cannot guess the data or the model, since they are encrypted when transferred. This approach has been tested for the case that one of the parties of the Federated Learning system is malicious and tries to attack the other parties by either trying to infer their data [27] or inserting a backdoor to learn their data [28].

Utilising Differential Privacy or Homomorphic Encryption are two different approaches on how Federated Learning can be extended to improve data privacy. Federated Learning is implemented in this work by utilising the python library Flower [29]. How this privacy can be compromised by attacking the model is described in the following Section.

2.2 Adversarial Attacks in Machine Learning

Adversarial machine learning aims to protect the machine learning pipeline to ensure its safety at training, testing, and inference time. Several attack types and defences have been invented over the years to get closer to this goal. The main types of adversarial attacks are Evasion, Poisoning, Inference and Extraction attacks. In order to give an overview over the spectrum of how a machine learning pipeline can be attacked, attack characteristics are described. Then, membership inference attacks are outlined in more detail alongside the attack method used in this work.

2.2.1 Attack characteristics

Adversarial attacks can be characterised in three dimensions [30, 31, 32, 33]

1. Influence of the attacker: train time vs test time
2. Goal of the attacker: targeted vs untargeted
3. Knowledge of the model: black-box vs white-box vs grey-box

The first dimension describes which part of the machine learning pipeline the attacker wants to influence. An attack can occur during training time, where the attacker tries to manipulate the training data and therefore influence the training process. For this, some knowledge of the

training data is assumed. This kind of attack is also called a **data poisoning attack**. The goal of these attacks is to corrupt the model by influencing its training process, leading to incorrect or biased outputs. Poisoning attacks can be particularly damaging, as they can degrade model performance or cause the model to behave in a controlled manner favourable to the attacker. [31, 32]

Conversely, attacks during test time do not influence the training process and rather manipulate data used at test time. Attacks at test time are denoted as **inference or exploratory attacks**. They do not try to alter the training process or its data, but rather try to discover information about the machine learning model or its training data. Therefore, this attack is more related to the data privacy field. A data privacy promise can be compromised by leaking information about the training data to the adversary using this attack. One attack method, namely the membership inference attack with the shadow model, is described in Section 2.2.2. [32]

The second dimension to characterise adversarial attacks is the goal of the attacker. They can either focus on one input, which is a targeted attack or on several inputs, an untargeted attack. Targeted attacks can be interpreted as changing the model to misclassify an example as a specific class. Whereas an untargeted attack aims at inducing a random misclassification.

As a final dimension, adversarial attacks have no knowledge of neither the dataset nor the model parameters or architecture. As the attacker does not have any knowledge about the model, it is called a black-box attack. This limited knowledge requires the attacker to use techniques, where they approximate the target model's behaviour using surrogate models. The attacker relies on the transferability of the inferences made from the surrogate models to the target model. Therefore, one goal for the attacker could be the extraction of the model by approximation to discover weaknesses in order to use them in an exploit. In the real world, this could be equated to a scenario where a bank is more prone to giving out loans to people who have fewer credit cards. Therefore, an adversary who discovers this as a threshold in the model, can use it as a weakness. They could cancel credit cards before the application for a loan and re-establish them afterwards, just to get a higher credit score. However, if the attacker has complete knowledge of the model's architecture and parameters, the attack is called a white-box attack. Additionally, a combination of these two scenarios is called a grey-box attack, if the attacker knows only parts of the model, e.g. only the architecture and not the parameters. [31]

2.2.2 Membership Inference Attack with the Shadow Model technique

Membership inference attacks (MIAs) are a type of adversarial attack that focuses on determining whether a particular data point was used in the training set of a machine learning model. These attacks pose significant privacy risks, especially in scenarios where sensitive data is involved. The goal of the attacker is to infer the presence of a specific individual's data within the training set, which can lead to severe privacy breaches.

The concept of MIAs was introduced by Shokri et al. [16], who proposed a shadow model technique to perform these attacks. The shadow model technique involves training multiple models (shadow models) that imitate the behaviour of the target model. The steps for conducting a membership inference attack using the shadow model technique as visualised in Figure 2.2 are as follows:

1. Shadow Model Training: The attacker creates several shadow models using a dataset that is similar but not identical to the target model's training data. These models are trained to mimic the target model's behaviour. For this, the dataset is split into member and non-member parts, where the member part is used to train the shadow models.

2. Attack Model Training: The attacker trains an attack model on the outputs of the shadow models. Here, an attack dataset is formed by combining the features of the dataset and the predictions of the shadow models into one feature set. The labels of this dataset are determined whether the shadow models were trained on the sample (member) or not (non-member). Therefore, the attack model learns to distinguish whether a given data point was part of the shadow models' training sets or not.

3. Inference: In the last step, the trained attack model is used to determine whether specific data points were part of the target model's training set. Usually, an attacker does not have access to the original train-test-split of the target model, therefore this data has to be guessed, e.g. by sampling from the distribution of the attack dataset.

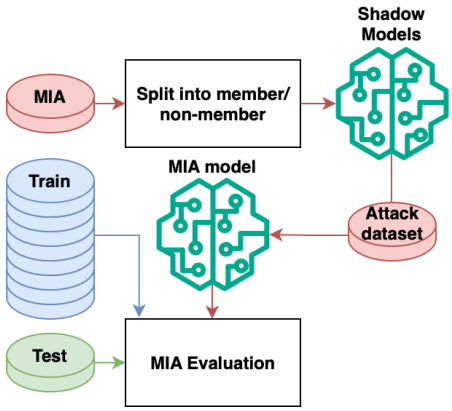


Figure 2.2: MIA with the shadow model technique

This shadow model technique for MIAs is implemented in this work with the Adversarial Robustness Toolbox for python [34].

Evaluation of Membership Inference Attacks

The effectiveness of MIAs is evaluated based on the attack success rate, which measures how accurately the attack model can determine membership status. Several factors influence the success of MIAs, including the Complexity of the target model, the similarity between the shadow models' training data and the target model's training data, and the overall structure of the attack model. Quantifying the success rate of MIAs is usually done by measuring the precision and recall (see Section 2.0.1) of the trained attack model used as a predictor for memberships status.

Membership inference attacks highlight the critical need for robust privacy-preserving techniques in machine learning. The development and implementation of methods like Differential Privacy and Federated Learning are crucial to mitigating these risks, ensuring that sensitive data remains protected while maintaining the performance and explainability of the models.

By understanding the mechanisms and implications of MIAs, this work aims to explore how privacy-preserving techniques can balance the trade-off between model explainability and data privacy, offering insights into developing secure and interpretable machine learning models. This aspect of explainability is covered in the next section.

2.3 Explainable AI

When looking at predictions of a model, it is not always clear how exactly the model came to that prediction. An explanation would help for either understanding the decisions of the model better or explaining the results to a user. Prediction models that are learned with data usually do not explain their results. Only if the predictors are simple enough themselves (e.g. linear or logistic regression, decision trees), they can be interpreted without using additional tools. However, simplicity often comes with drawbacks of lower prediction performance. Therefore, tools were developed to make the results of complex, learned models interpretable or explainable. These tools are summarised under the term explainable AI (XAI).

In XAI, the methods for interpreting the results of a model have been categorised in the book Interpretable Machine Learning by Christoph Molnar [35]. These categories are widely used in literature, as a survey by Burkart et al. [36] underlines. XAI methods are therefore split into

1. intrinsic vs. post-hoc
2. model-specific vs. model-agnostic
3. local vs. global

As mentioned before, a prediction model can be restricted in their Complexity in order to be interpretable by itself, i.e. the model is intrinsically interpretable. However, if a model is too complex, methods can be used to analyse the model after the training finished (post-hoc).

The second criteria for distinguishing XAI methods is the difference between model-specific and model-agnostic. If an XAI method can be used for multiple prediction model classes, it is model-agnostic. This attribute has the natural upside that a model-agnostic XAI method can be used for different prediction models. However, by being more generally applicable, the method can be less accurate in interpreting the results. Consequently, if an XAI method is model-specific and can only be used for a specific model class, it can provide more precise explanations.

As a last criterion, XAI methods can be categorised in giving local or global explanations. Local explanations are given by methods that either explain an individual prediction or only a part of the data. With this focus on a specific area, interpretations can be more tailored to these samples. If an XAI method should give an interpretation of the behaviour of the entire model, it is called a global method.

In order to answer the research questions of this work, local, model-agnostic, post-hoc XAI methods are used. Investigating models that can be explained intrinsically would be preferable, but is often not realistic for complex use cases. By selecting model-agnostic XAI methods, a more generalised conclusion can be drawn, which can then be applicable to other domains than the health domain. Finally, local XAI methods were investigated, since they would provide more meaningful explanations for each sample in our dataset which is representing a patient. Similarly to a doctor or a nurse, the local explanations would give reasons as to why the prediction was made for this specific patient. This decision is based on preference, as it seems more natural to investigate explanations that have similarities to talking to a doctor or nurse.

The XAI methods and metrics used in this work are implemented with the python library Quantus [37].

In the following sections, the selected XAI methods are described and then their evaluation process is presented.

2.3.1 Methods

For interpreting the results of the experiment models, Local Interpretable Model-agnostic Explanations (LIME) [38], Integrated Gradients [39] and Feature Permutation [40] are utilised as

they are widely used in literature. Additionally, they cover different XAI approaches, namely training a local surrogate model (LIME), utilising the gradients of the trained model (Integrated Gradients) and altering input data and examining changes in the prediction output (Feature Permutation). They are model-agnostic XAI methods. Therefore, they are suitable for examining the trade-off between explainability and susceptibility to adversarial attacks on a higher level, where the specific model choice is not of high importance. Each of these methods return an attribution vector, that quantifies the importance of each feature to the prediction of the model [36]. The three named attribution methods are described below.

LIME

LIME [38] aims towards giving interpretable explanations for predictions given by a local model. This is achieved by training a simpler surrogate model on data points, with their importance weighed by the distance to the sample that is to be explained. A measure of fidelity and Complexity is chosen to select an interpretable local model that is both simple and closely approximates the predictions of the predictor being explained. One challenge for applying LIME is the correct definition of the proximity the surrogate model should approximate [35]. This can be formalised as in Def. 5.

Definition 5 *Let $g \in G$ be an interpretable model of a family of interpretable models G (e.g. linear models), $\mathcal{L}$ be a measure of fidelity, Ω be a measure of Complexity, f be a model to be explained at sample x in the proximity π_x, then the attribution ξ of sample x is defined as*

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}}\, \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Through this definition, the attributions $\xi(x)$ are given by a surrogate model that closely estimates the predictions of the model being explained in the vicinity of a sample x.

Integrated Gradients

Intuitively, Integrated Gradients [39] determines the total contribution for each feature by measuring the size of the impact on the function by stepping from a baseline to the sample that is currently being evaluated. By using a baseline as a reference, this method naturally contextualises resulting attributions.

With dataset $X \subseteq \mathbb{R}^n$ and selected baseline $x' \in \mathbb{R}^n$, Integrated Gradients is calculating the integral along the path from the baseline x' to the sample $x \in X$ that is to be explained. For dimension $i \in \{0, 1, ..., n\}$, the attribution values for this method are defined as in Eq. 2.9.

$$IntegratedGrads_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha \tag{2.9}$$

Therefore, each feature at dimension i of the input sample x is attributed a value of importance with regard to the baseline x'.

Feature Permutation

Measuring the difference in the model error by permuting features in the data is called the method of Feature Permutation [40]. As the name suggest, for each feature the input sample is changed and the difference or quotient of the original measured error and the error of the permuted sample is calculated. This is called the feature importance, which the features then

get ranked by. Formally, the feature importance FI_i for each feature at dimension i of the input sample x, its permutation x', with error function e is defined as shown in Eq. 2.10.

$$FI_i(x) = e(x') - e(x) \text{ or } FI_i(x) = \frac{e(x')}{e(x)} \tag{2.10}$$

Any one of the two methods to calculate feature importance can be chosen to rank the features by their impact on the prediction outcome.

2.3.2 Metrics

In the field of Explainable Artificial Intelligence, it is crucial to assess how well explanations provided by XAI methods fulfil their intended purpose. This section discusses metrics for evaluating XAI explanations selected for this work, focusing on a broad range of properties of explanations, namely Faithfulness, Robustness, Complexity, Sensitivity and two metrics testing axiomatic properties. Each of these properties, and their specific metrics used, are described and explained in the following.

Faithfulness

Faithfulness describes how well the explanation approximates the prediction of the model. High Faithfulness indicates that the explanation is a good surrogate for the model's actual decision-making process. Some models might only achieve local Faithfulness, meaning they accurately explain predictions for specific data subsets rather than globally.

The **Faithfulness Estimate** [41] is measuring the probability drop of the predicted class when obscuring or removing features. Comparing this drop in probability to the feature relevance prediction of the XAI method, then forms a correlation of feature importance and model prediction behaviour.

Monotonicity [42] determines if the class probability monotonically increases with starting from a baseline and replacing values of the baseline with actual values of features, starting from least relevant to most relevant. If the features are correlated with the prediction, then the probabilities should increase monotonically.

Sufficiency [43] evaluates whether an explanation used to justify a prediction for one sample can reliably predict the same prediction for any other sample with the same property. This metric ensures that the explanation is consistent and reliable across different instances.

Robustness

Robustness metrics measure the stability of explanations when the input sample is slightly altered. The expectation for a good explanation method is that the attributions should remain similar to the original explanations, which were given for the unaltered input.

Average and Maximum Sensitivity [44] measures the maximum or average difference between explanations in a predetermined radius around the investigated sample. Points are sampled within this radius using a Monte-Carlo approach, to estimate the true maximum or average Sensitivity.

Local Lipschitz Estimate [45] samples around the input to find a point of the greatest ratio between the difference of explanations and the difference of samples. With this point, a constant can be determined which satisfies the local Lipschitz criterion. This constant then represents how different the explanations are within a small radius of the original sample are. As there is no single desirable value for this constant, the resulting value is always specific to the applied

domain and explanation goals. Hence, this method is used to compare XAI methods within a specific domain.

Complexity

Comprehensibility evaluates how easily humans can understand the explanations. While comprehensibility is essential, it is often gauged through user studies, which are beyond the scope of this work. Hence, Complexity metrics can serve as a proxy. To measure the Complexity of explanations, one investigates how concise they are, i.e. how many features are deemed important by the XAI method.

Complexity [46] is defined as the entropy of a probability distribution, which describes how important features are in comparison to all other features. An explanation with high Complexity would then also be evaluated with a high entropy value.

Effective Complexity [47] is calculated by ordering the features of a sample by importance and giving the amount of relevant features necessary that the expected loss value of the prediction model is within a certain threshold of $\epsilon > 0$. Therefore, an explanation with low Effective Complexity is interpreted as simple and broad, because features with minor importance are attributed a negligible effect.

The **Sparseness** [48] of an attribution vector is measured by the Gini Index of the absolute values of the vector. Before calculating the Index, the absolute values are sorted from biggest to smallest. When calculating the Gini Index, a higher Sparseness and thus lower Complexity of the explanation is given by a larger Gini Index value.

Sensitivity

Sensitivity assesses the extent to which explanations differ between models trained on the same task but possibly using different features. Sensitivity is crucial for reliability, but achieving it can be challenging, as different models may arrive at the same prediction using varied pathways. By evaluating how XAI methods react to changes of their inputs, the Sensitivity of this method to these changes is tested. A selection of metrics quantifying Sensitivity is described in this section. Hereby, Sensitivity differs to Robustness in that it is focused more on the alteration of the machine learning model, e.g. the model parameters.

The **MPRT (Model Parameter Randomisation Test)** [49] quantifies Sensitivity by randomising model layers in a cascading fashion or by only randomising single layers and keeping the rest fixed. Then, the correlation between the explanation of this altered model and the attributions of the original model are measured. The underlying idea is that since alterations to the model parameters have a strong effect on test performance, an XAI method should be affected similarly.

Efficient MPRT [50] re-interprets the original MPRT test by measuring the difference in Complexity before and after all the parameters of the prediction model have been randomised. The difference in Complexity is measured by calculating the relative increase of entropy from the original explanation to the one of the fully randomised model.

Smooth MPRT [50] reduces the impact of noise on the original MPRT evaluation results through sampling. MPRT is extended by adding a preprocessing step, where the input sample is perturbed into multiple instances and their calculated attributions are averaged. Afterwards, the original MPRT is continued.

Axiomatic

Now follow descriptions of XAI metrics that evaluate whether or how well an XAI method follows predefined axiomatic properties.

Completeness [39] determines whether the sum of attributions given by the XAI method is equal to the difference between the outputs of the prediction model for an input sample and a previously set baseline. This is a validation method for checking the comprehensiveness of the explanations given by an XAI method.

Non-Sensitivity [47] is quantified by determining the cardinality of the symmetric difference of the set representing features with an attribution of zero and the set representing the features which the prediction model is not functionally dependent on. Hence, measuring how accurately the XAI method deems features as unimportant which the prediction model is also not dependent on.

2.4 Related Work

This section reviews the existing literature on privacy-preserving techniques with a focus on Differential Privacy and Federated Learning. Furthermore, it explores, explainable AI (XAI) and the intersections between XAI and Differential Privacy and Federated Learning with a focus on the explainability-privacy trade-off and Membership Inference Attacks.

2.4.1 Overview on Privacy-Preserving Techniques

Techniques that preserve the privacy of the data used have been extensively developed and investigated. Three main areas of interest have been emerging: Privatising the training data by adding noise and altering the data, e.g. with Differential Privacy [51, 52, 53, 8], Federated Learning [51, 54, 55, 56] so that the data does not have to leave the data owners and secure communication through homomorphic encryption [57, 58, 59, 60] in a collaborative learning setting. They all serve the purpose of upholding the data privacy promise with different approaches, and all of them can be combined [61].

Adding noise to the training data usually has drawbacks of lowering the utility of machine learning models, but might be necessary in order to reduce their vulnerability of reconstructing the data [8, 52]. However, these negative effects can be mitigated by using a Federated Learning approach [9]. Federated Learning in itself already serves a privacy purpose, since in a collaborative learning setting only the gradients are shared from the collaborating partners with a central server. Hence, the training data each participant uses to train a model is not distributed and is therefore kept from other partners in this setting. Nevertheless, if the data is not privatised during training, the data can still be inferred from the model itself [4]. Consequently, the gradients can be encrypted before sharing them with the server where they are aggregated, sent back and then decrypted to update the local models [57, 58, 59, 60]. With this method, communication between the clients and the server is more secure and can prevent data leakage through the training pipeline in a collaborative setting.

Since homomorphic methods put their focus on the communication between clients and the central server, it is considered out of the scope of this work. Therefore, only the widely spread techniques of adding noise to the data by applying the Differential Privacy concept and Federated Learning are considered and presented in more detail.

2.4.2 Federated Learning combined with Differential Privacy

Federated Learning in combination with Differential Privacy can be effective to improve the privacy of data. Federated Learning allows multiple clients to collaboratively train a model without sharing their raw data, thus already enhancing privacy. The value of combining a collaborative learning approach with Differential Privacy has been shown by many studies. Shokri et al. [51] highlighted this value by utilising Differential Privacy and only sharing gradients partly with the server. The privatised models achieved similar accuracy to normal non-distributed models which showcases the potential of these methods.

Furthermore, Geyer et al. [10] introduced a client-side Differential Privacy mechanism into Federated Learning to prevent the revelation of individual client participation during training. Their algorithm achieves client-level Differential Privacy with minimal performance loss and suggests dynamically adapting Differential Privacy during Federated Learning to improve model performance, as already suggested e.g. in [62].

The combination of Federated Learning and Differential Privacy can even be extended by also applying homomorphic encryption, which improves privacy of communication. This approach was explored by Hao et al. [61], where the privatised model only lost little accuracy compared to the non-private model and therefore emphasising the impact of combining techniques. It was even demonstrated that the combination of Federated Learning with Differential Privacy can mitigate negative effects of Differential Privacy [9] further indicating the potential of combining these techniques.

2.4.3 Explainability of Differential Privacy and Federated Learning

Differential Privacy is a well-established technique that ensures the privacy of individual data entries by introducing noise into the data or model parameters. Patel et al. [63] developed a framework for generating Differential Privacy compliant, model-agnostic post-hoc explanations for black-box machine learning algorithms. Their work focuses on maintaining the quality of explanations while protecting sensitive information, demonstrating that private explanations leak significantly less information compared to model predictions. They propose an adaptive Differential Privacy explanation algorithm to better approximate the underlying decision model of black-box algorithms. This work further establishes the use of protecting data with Differential Privacy while still upholding a certain degree of explainability. In contrast to this work, they did not investigate the interplay of Differential Privacy and Federated Learning and focused on Differential Privacy.

When using Differential Privacy, noise is added to the data, therefore it can be difficult to maintain explainability. Saifullah et al. [9] recently proposed a benchmark for integrating Differential Privacy with XAI methods. They highlight that while Differential Privacy typically reduces the interpretability and fidelity of XAI heatmaps for image data, Federated Learning can mitigate these negative impacts. They even recommend optimising Federated Learning before applying Differential Privacy when they are used in combination, as Federated Learning counters some of the problems induced by Differential Privacy. This emphasizes the power of the interplay of Differential Privacy and Federated Learning not only by the effect each of them have on data privacy, but also the outcome of their combination. Furthermore, they suggest that perturbation-based XAI methods should be used in applications requiring both privacy and explainability, because they are less affected by using Differential Privacy during training. They did investigate the interplay of Differential Privacy and Federated Learning, but they did not investigate the practical implications of privatising the models, e.g. through attacking it with an actual adversarial attack.

2.4.4 Explainable AI and Adversarial Attacks

Explanations generated by XAI methods can be exploited for adversarial attacks, as shown by Kuppa et al. [12]. They discuss how adversarial attacks including model extraction, adversarial attacks, membership inference, and poisoning attacks can exploit the explanations of a model by using counterfactuals. They propose a black-box attack that leverages XAI methods providing counterfactuals to compromise the confidentiality and privacy properties of machine learning classifiers by utilising the counterfactuals as training data for a shadow model imitating the target model. Their attack methodology, tested on three cybersecurity-relevant datasets, shows superior performance compared to state-of-the-art methods under realistic threat models. This shows the need for more research on how to effectively protect data from attackers.

They used the concept of Membership Inference Attacks by using a shadow training technique, first introduced by Shokri et al. [16]. They quantify the information leakage through model prediction outputs and demonstrate that machine learning models, including those offered by machine-learning-as-a-service platforms, can reveal significant information about their training sets. Their study emphasizes the need for robust privacy-preserving mechanisms to mitigate such risks.

2.4.5 Evaluation of XAI methods

In order to compare the quality of different explanations given by XAI methods, these explanations have to be either investigated through a user study or by measuring with objective metrics. This work focuses on objective metrics to efficiently determine the quality of explanations and providing easier comparability to other research.

A comprehensive survey by Vilone and Longo [64] provides a broad overview on how XAI methods are evaluated by metrics that are exploring different aspects of the explanations. This survey offers foundational insights into the current state of research on how XAI methods can be compared with each other.

Furthermore, Hedström et al. [11] investigate two complementary performance characteristics of quality estimators on image data in XAI: resilience to noise and reactivity to randomness. Their study on image data utilises various XAI metrics, including Complexity, Faithfulness, localization, randomization, and Robustness, to evaluate the quality of explanations. This meta-evaluation approach circumvents the need for ground truth labels and provides a comprehensive framework for assessing XAI methods. This is a great example on how to evaluate explanations of a machine learning model on a meta level.

Jyoti et al. [65] provide a survey on the Robustness of XAI methods, exploring how different XAI techniques withstand various perturbations and adversarial conditions. This survey complements the work by Hedström et al., offering additional perspectives on the resilience and reliability of XAI methods.

2.4.6 Summary

The literature reveals a complex interplay between Differential Privacy, Federated Learning, and explainable AI, with significant implications for the explainability-privacy trade-off and the susceptibility of models to adversarial attacks. While Differential Privacy and Federated Learning offer robust privacy guarantees, their integration with XAI poses challenges in maintaining interpretability and model performance. Existing studies provide a foundation for this work, which aims to further explore and quantify this trade-off, offering practical recommendations for deploying privacy-preserving and interpretable machine learning models in real-world applications.

Chapter 3

Approach

In order to answer the research questions mentioned in Chapter 1.1, first the problem has to be analysed and properly defined. Therefore, the problem is first described before a solution is proposed. Alongside, experiments that are conducted to test the solution approach, are described.

3.1 Problem description

In order to investigate the problem, it is necessary to first define it and give reason as to why the research questions are asked in Chapter 1.1. Primarily, this work is investigating an explainability-privacy trade-off where a machine learning model should be trained such that it is both explainable while also guaranteeing data privacy. Therefore, both the explainability and the ability to keep a data privacy promise both have to be investigated and tested. In essence, there are two separate parts that have to be explored with this problem. The results of each part should then be combined to find how explainability and data privacy are connected:

1. **Explainability**: How is the explainability of a model affected when privacy-preserving techniques are applied?
2. **Data privacy**: How well is the original data protected from an adversary that is trying to extract the original data from the machine learning model?

In order to examine the first part, machine learning models have to be trained with multiple privacy-preserving techniques. The results of these different techniques should then be analysed and compared to determine which of them can achieve good explainability of the model while privatising the data. Furthermore, to inspect the second part, an attack on these trained models has to be simulated. With this attack, it can be measured how well the data is kept private.

From this description of the problem, a solution that is investigated in this work is outlined.

3.2 Proposed Solution

To explore this problem, the approach followed in this work involves a systematic analysis of the trade-offs between explainability and privacy by utilising privacy-preserving techniques and evaluating their impacts. The machine learning model will be assigned a certain privacy budget and number of clients, and the quality of explanations provided by different XAI methods will be assessed. The proposed solution consists of the following steps:

Training Models with Privacy-Preserving Techniques Machine learning models will be trained using Differential Privacy and Federated Learning. Various levels of privacy budget will be assigned in Differential Privacy to understand its impact on model performance and explainability. Federated Learning will be used with different number of clients to evaluate their effects.

Evaluating Explainability The quality of explanations generated by XAI methods listed in Chapter 2.3.1 will be assessed. Metrics outlined in Chapter 2.3.2 grouped by faithfulness, robustness, complexity, sensitivity, and axiomatic will be used to measure the impact of privacy-preserving techniques on explainability.

Simulating Adversarial Attacks Membership Inference Attacks will be simulated to evaluate the privacy guarantees provided by Differential Privacy and Federated Learning. The success rate of these attacks will be measured to determine how well the data is protected.

Analysing the trade-offs The results of the aforementioned steps will be analysed to understand the trade-offs between explainability and privacy. This analysis will provide insights into the effectiveness of combining Differential Privacy and Federated Learning, and offer guidelines for selecting appropriate privacy-preserving techniques based on specific application requirements.

The first two steps are used to answer RQ 1 and therefore investigating the effect of DP and FL on the explainability of a classification model. Furthermore, simulating adversarial attacks and then analysing the trade-off between explainability and privacy is used to answer RQ 2.

By following this solution approach, this research aims to provide a thorough evaluation of the explainability-privacy trade-off in machine learning models employing Differential Privacy and Federated Learning. The ultimate goal is to offer actionable insights for practitioners seeking to deploy privacy-preserving and interpretable machine learning models in real-world applications. The technical setup of this proposed solution is now described in the following section.

3.3 Experimental Setup

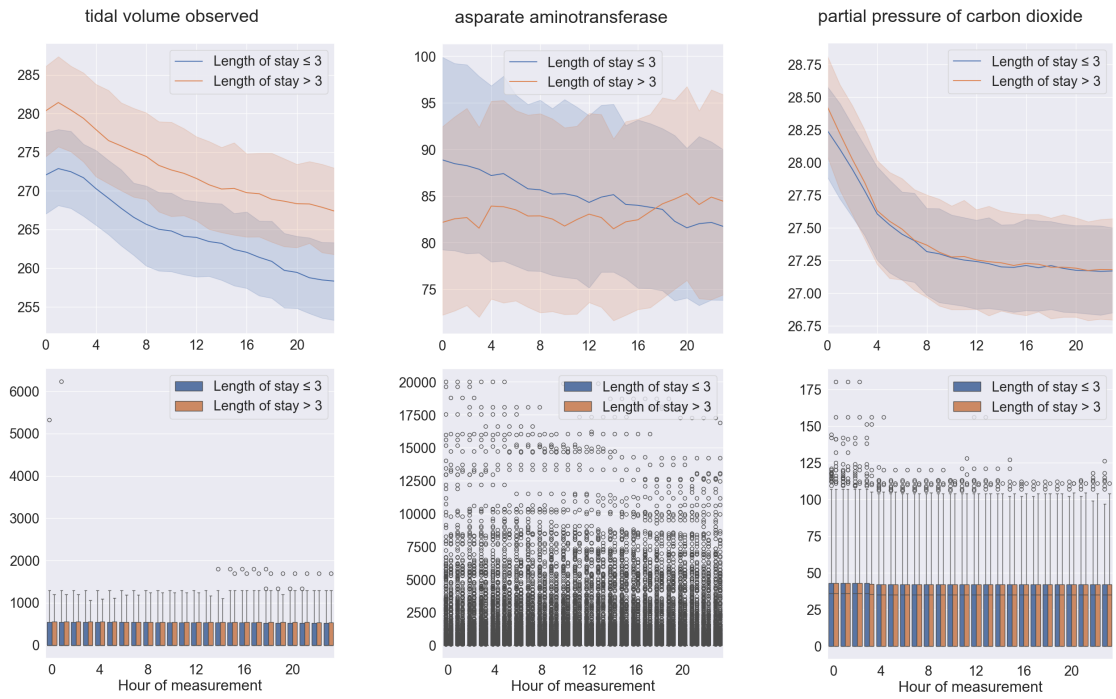
The experimental setup is designed to systematically investigate the trade-offs between explainability and data privacy in machine learning models using privacy-preserving techniques. The following steps outline the approach:

3.3.1 Dataset and Cross-Validation

For the experiments conducted, many datasets could have been chosen. In order to stay close to a realistic scenario where the risk of leaking data should be minimised, a medical dataset is selected. Since there are regulatory security standards for electronic health data, e.g. the GDPR [5] in the EU and the HIPAA [7] in the U.S., this work investigates this domain.

For this investigation, the Medical Information Mart for Intensive Care (MIMIC) [66] dataset is chosen. As this dataset is publicly available and often used, there are many benchmarks to validate the training results of the experiments. MIMIC contains deidentified patient data from critical care units. The availability of data that is already cleaned from outliers and missing data for this dataset makes it even more attractive as it is already prepared for training. One example of data prepared for machine learning tasks and used in this work is the MIMIC-Extract

[67]. It provides both cleaned data and a benchmark for several classification tasks. MIMIC-Extract transformed vital sign and laboratory measurements from the third version of MIMIC into hourly time series data. The transformation includes unit conversion, outlier handling and the aggregation of semantically similar features. The transformed data is used in this work to compare training results to one of the benchmark test that were performed in the MIMIC-Extract paper, namely length of stay prediction.



(a) Tidal volume observed (b) Aspartate aminotransferase (c) Partial pressure of CO_2

Figure 3.1: The values of three features (tidal volume observed, aspartate aminotransferase, partial pressure of CO_2) resulting from the MIMIC-Extract pipeline. Sample are aggregated by their respective length of stay classification. The upper plot of each feature displays the hourly average feature values with a confidence interval of 0.95. The lower plot each shows the distribution of the values at each hour.

In total, all 68 distinct medical measures were used from the dataset, each with hourly measurements of the past 24 hours combined with a mask if the values were imputed and how long ago the last real measurement was taken as provided from the original dataset. This results in a total feature space of 4896 input features. A selected three hourly measurements are visualised in Figure 3.1. By visualising these features, one can see that for one, there are noticeable differences between the classes (e.g. Figure 3.1 a), but also that there are a lot of outliers for samples belonging to either class. Many outliers would be expected, as the hope is that through these outliers a doctor or a nurse can determine what the patients need or how to diagnose them.

A 10-fold cross-validation approach is employed to ensure robust and reliable results. During cross-validation, one fold is set aside for testing the performance of the model as usual and one fold is reserved for conducting a Membership Inference Attack to train the shadow models.

3.3.2 Target Model

With the data as detailed before, each model is trained to be evaluated on performance, explainability and being targeted by a Membership Inference Attack. For this, the models are trained to classify the data for a length-of-stay-prediction problem. They should classify whether a patient is staying three days or longer at the hospital or if they are staying shorter than three days.

In order to validate the training process, a logistic regression (LR) model is used to compare to the results of the MIMIC-Extract benchmarks [67]. However, a linear model is hardly a black-box model and therefore easier to explain by XAI methods. Therefore, not only LR is used in this work, but also a simple neural network. The logistic model is optimised with gradient descent [68] throughout the experiments of this work. For the neural network, a feedforward network [68], that contains two hidden layers with 32 nodes each and leaky ReLU [69] activations, is utilised. The network is optimized with the AdamW [70] optimisation algorithm. For the single output node, the Sigmoid activation function is used.

To optimise the models for training, the cross entropy loss is utilised alongside stochastic gradient descent for Logistic Regression and the AdamW algorithm [70] for the neural model as optimisers. Additionally, the hyperparameters (e.g. learning rate, training epochs, neural net size) are empirically adjusted for better performance. Both models are optimised with a learning rate of 0.001 over 100 epochs. In the federated learning setting, 100 global training steps were taken with 5 local optimisation steps.

Experiments conducted

An overview of the experiment pipeline is visualised by Figure 3.2. As already discussed, for each cross-validation step the data is split into 10 folds, where one is used as a test set and one fold is used as data for the shadow models during the attack. The training evaluation of the target model is used as a benchmark to compare to previous work and analyse differences between different methods used in training. How this pipeline is used to answer the research questions asked in Chapter 1.1, is outlined in the following sections.

For easier readability, the research questions are repeated here:

1. What is the effect of applying Differential Privacy (DP) and/or Federated Learning (FL) on the explainability of a classification model?

2. To what extent can the explainability of models trained with DP and/or FL be preserved, while concurrently minimising the success rate of external attackers attempting to infer information about the training data through a black-box membership inference attack?

Research Question 1: Explainability with DP and FL For investigating the effect of Differential Privacy and Federated Learning on model explainability, each model is trained with each combination of different privacy budgets and number of clients for Federated Learning. To analyse the models with different ϵ-differential privacy constraints, multiple values for ϵ have been selected with $\epsilon \in \{0.5, 1.0, 3.0\}$. Furthermore, a model is optimised with no application of Differential Privacy for comparison. In order to conform to the library used and to enable a faster runtime of the experiments, the ϵ-differential privacy budget is relaxed with $\delta = 0.01$. For Federated Learning, client numbers 5, 10, 20 and 50 are explored. Additionally, a centralised model is evaluated that not only the combination of Differential Privacy and Federated Learning is examined, but also a baseline to further visualise the effects of each privacy-preserving technique.

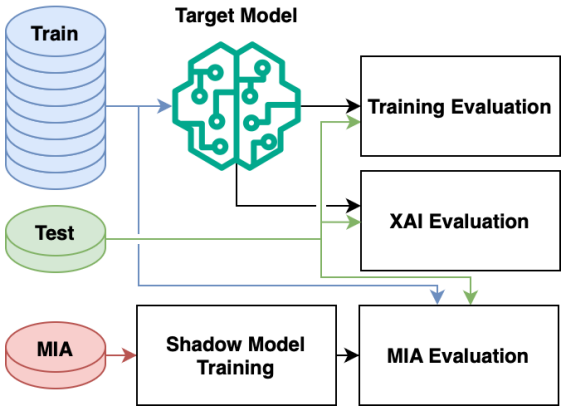


Figure 3.2: Overview of the experiment pipeline

Post-training, the results will be explained using the following Explainable AI methods that are presented in Chapter 2.3.1, namely Integrated Gradients, LIME and Feature Permutation. Herein, only 50 samples are randomly selected to allow efficient experiments and still provide a good estimate of the evaluation as if the entire test set is used. These explanations will then be evaluated using the metrics listed in Chapter 2.3.2 grouped by faithfulness, robustness, complexity, sensitivity, and axiomatic properties. For the XAI assessment, the target model is evaluated on the test data as depicted in Figure 3.2.

These evaluations are giving vital insights into the interplay between explainability and privacy of a model and provide an understanding of the explainability-privacy trade-off.

Research Question 2: Privacy Evaluation with MIA For answering the second research question stated in Chapter 1.1, the models trained with different combinations of utilising Differential Privacy and Federated Learning as described previously, are now assessed on their effectiveness in keeping the training data private. This assessment is conducted by measuring the success rate of a membership inference attack using the shadow model technique as described in Chapter 2.2.2. Therefore, 3 shadow models are trained on the fold set aside for this attack with a global 50/50 train-test-split shared with each shadow model. By training them with this data, these shadow models try to imitate the target model. Hence, their architecture is selected depending on the target model. For neural models, the same neural architecture is used for the shadow models. Similarly, logistic models are used for the logistic target models.

The performance of these models is assessed with the test set by comparing it to the performance of the target models. This step is crucial to ensure similar behaviour of the shadow models. Finally, new data is generated by gathering the predictions of the shadow models over the entire MIA fold. Here, the features and predictions become the new features for the attack model, whereas labels now determine whether the samples have been used for training the shadow models or not. Consequently, the attacking model is another classification model that is trained on this newly generated data. The success of the attack is qualified by the precision and recall of membership predictions given by the attack model on the training and test data of the target model.

The neural model architecture features two distinct heads for processing input features and input labels of the generated shadow model data. The first head, dedicated to input features, consists of three fully connected layers (512, 100, 64 nodes respectively) with ReLU activations.

The second head, dedicated to input labels, includes two fully connected layers (256, 64 nodes respectively) with ReLU activations. These two head outputs are then concatenated and passed through a final layer, followed by a sigmoid activation function to produce a single output for membership prediction.

This setup ensures a deep analysis of the explainability-privacy trade-off by systematically training models with privacy-preserving techniques, evaluating their explainability, and simulating adversarial attacks to test data privacy. The results from these experiments will provide valuable insights into the effectiveness of combining Differential Privacy and Federated Learning in maintaining a balance between explainability and privacy.

Chapter 4

Results

In this chapter, the results of the experiments proposed in Chapter 3.3 are presented. First, the prediction performance results are compared with the length-of-stay prediction model of [67] in order to show that the results of this work are comparable and meaningful. Then, the evaluations of the two experiments are displayed and described before summarising all results. These results are then discussed in Section 5.1.

Models are displayed in groups of how many clients have trained on the data, or if a centralised model is used. In these groups, the ϵ privacy budget given by Differential Privacy is displayed. If Differential Privacy is not used, it is indicated with "n/a".

4.1 Model Prediction Performance

4.1.1 Comparison to related work

To show that the results presented in this work are valid and comparable, the performance of the logistic predictor is compared to the one used in [67]. Their result of the length-of-stay prediction evaluation is displayed in Table 4.1, where they achieved a performance scores of 0.716 for AUROC, 0.651 for Accuracy and 0.594 for the F1 Score.

	AUROC	Accuracy	F1 Score
LR model from [67]	0.716	0.651	0.594
LR baseline model from this work	0.655 ± 0.037	0.619 ± 0.03	0.569 ± 0.025

Table 4.1: Baseline comparison to MIMIC-Extract [67] with the average and standard deviation of the baseline logistic model of this work

As depicted in Table 4.1 the baseline with no Differential Privacy and a centralised model gives similar results to previous research. Whereas, the AUROC score of the baseline is slightly worse, with an average AUROC Score of 0.655. This can be attributed to the different libraries that were used to implement logistic regression, and therefore differences in the performance outcome can occur. Nevertheless, this emphasizes the validity and comparability of the results presented in this work.

4.1.2 Comparison of different model configurations

The performance results of the different configurations of each model are shown in Figure 4.1. To explore which parameter settings for Federated Learning and Differential Privacy are significantly better or worse than others, a pairwise statistical significance test was utilised, as explained in 2.0.2. The results of this significance test are visualised in Figure 4.2.

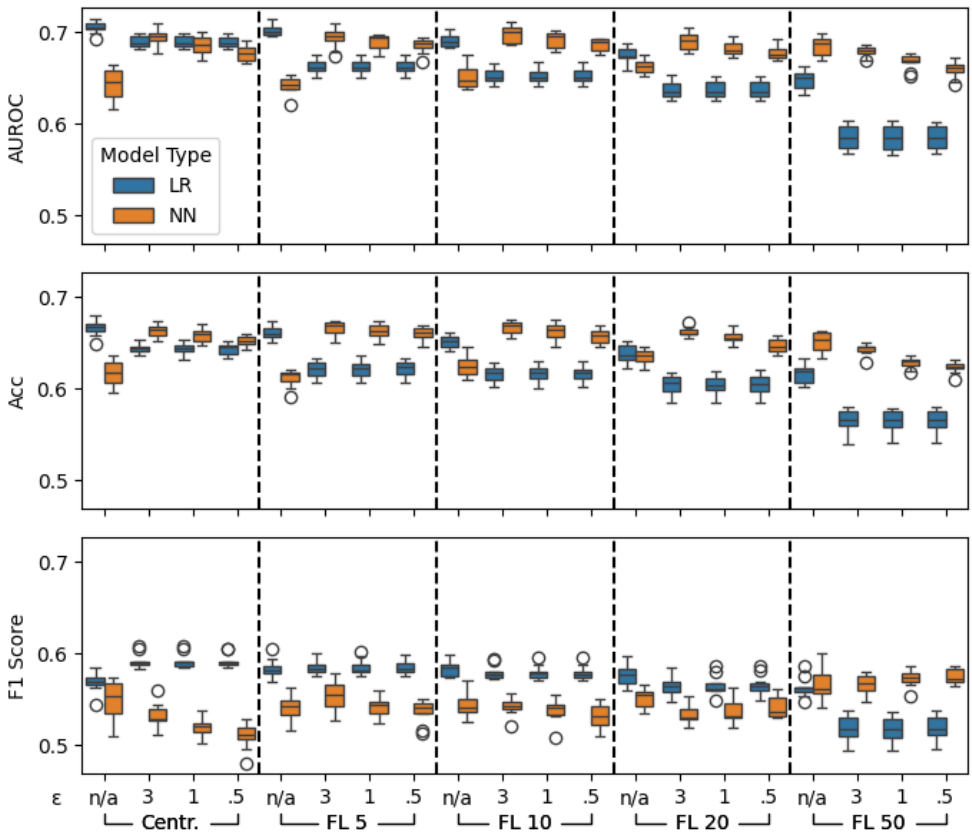
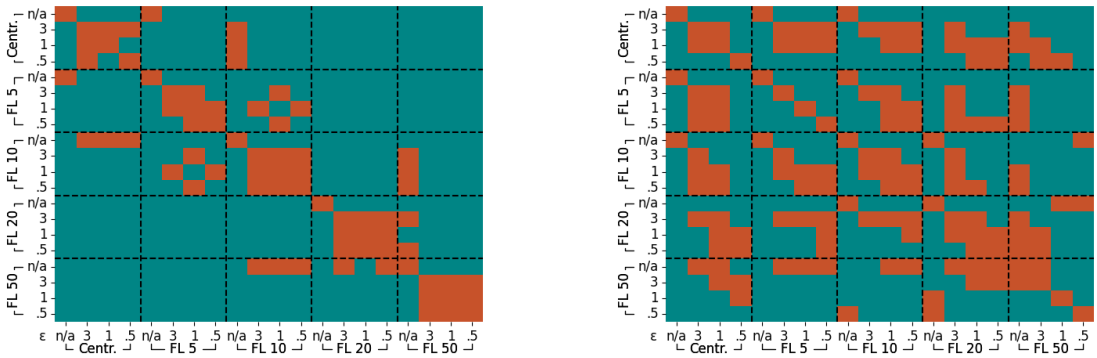


Figure 4.1: Prediction Performance (AUROC, Accuracy, F1 Score) of each model grouped by their model type. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

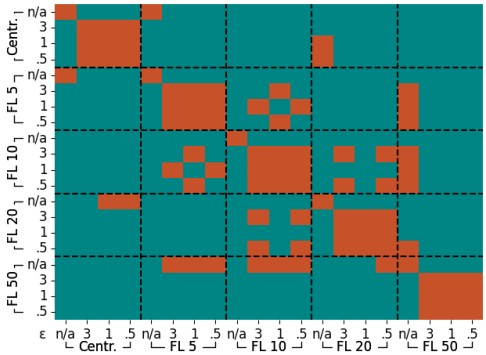
For the centralised neural model, the F1 score decreases with a lower privacy budget, however using Differential Privacy with the logistic model significantly increases the model performance. In contrast, the AUROC and accuracy scores of the centralised models are giving opposite results. Here, using Differential Privacy is beneficial for the neural model, but decreases the logistic model performance. Hence, with the dataset given, there is no clear general indication for the centralised models whether Differential Privacy is improving or decreasing the performance of a model. In this case, the performance results seem model dependant.

When looking at the evaluation scores for Federated Learning with 5, 10 and 20 clients, no clear trend can be observed for the F1 score as the results are very similar and not significantly

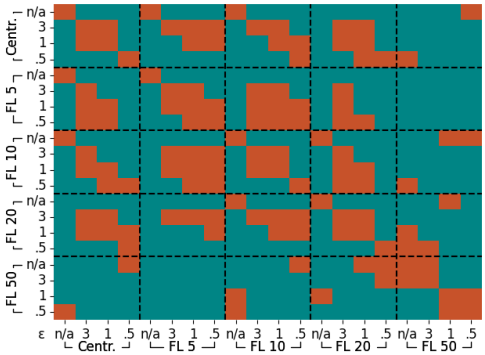


(a) LR AUROC

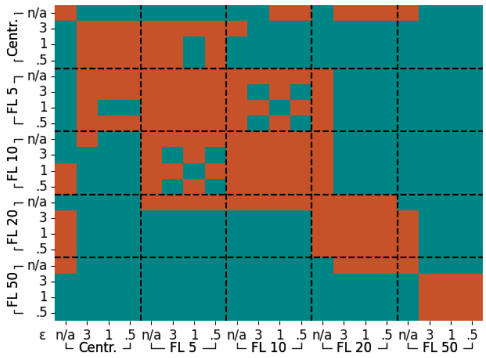
(b) NN AUROC



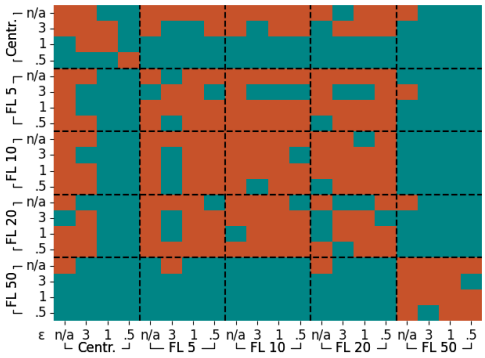
(c) LR Accuracy



(d) NN Accuracy



(e) LR F1 Score



(f) NN F1 Score

Figure 4.2: Student's paired t-test of each Performance metric (AUROC, Accuracy, F1 Score) of each model configuration for comparison with counterparts. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest, as the noise is increased in this order. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

different. Only Federated Learning for the logistic model with 20 clients in combination with Differential Privacy is significantly worse than the counterparts with the same privacy budget for Federated Learning with 5 and 10 clients. However, it is apparent that for the logistic model, similarly to the centralised models, the accuracy and AUROC score of Federated Learning with 5, 10 and 20 clients using Differential Privacy is significantly worse than the models not using Differential Privacy. Whereas for the neural model, using Differential Privacy is again beneficial and only decreases the AUROC and accuracy scores when a lower privacy budget is utilised. This further emphasizes the dependency of performance on the model type when utilising Differential Privacy.

As for Federated Learning, the number of clients does only have a general significant bad effect on most AUROC and Accuracy scores of the logistic model. Here, using more clients is disadvantageous. The only significant drop in the F1 score of the logistic model can be observed when using Differential Privacy with 50 clients, compared to not using a privacy budget. However, this negative effect cannot be observed with the neural 50 client model. Using 50 clients for Federated Learning with the neural model yields mostly significantly better F1 scores when comparing to the counterparts with the same privacy budgets. Overall, the worst logistic models for all Performance metrics are trained with 50 clients and Differential Privacy with any privacy budget. This indicates a limit for how many clients the dataset can be distributed to, before the performance suffers for the logistic model.

In summary, the privacy budget given by differential privacy has a bigger effect on the neural model than the logistic model and therefore is dependent on the model type for this experiment. In general, using Differential Privacy can have different effects of either improving or worsening model performance depending on the metric and model type. Using more clients for the federated models does not significantly change the performance of a model until a certain threshold.

4.2 Explainability evaluation of Differential Privacy and Federated Learning

For the evaluation of the explainability of the different model configurations, only LIME is considered in detail. Furthermore, the results of the evaluation of Integrated Gradients and Feature Permutation did not give additional insights to the general explainability of the models. The results of Integrated Gradients and Feature Permutation can be viewed in Appendix A.1.1 and A.1.2.

In this section, the XAI metrics as described in Section 2.3.2 are presented in their respective groups: Faithfulness, Robustness, Complexity, Sensitivity and Axiomatic.

4.2.1 Faithfulness

First, the results of the Faithfulness metrics Faithfulness Estimate, Monotonicity and Sufficiency are explored. For all XAI methods, the binary Monotonicity criterion was satisfied and did not change for different privacy budgets or number of clients. However, the Sufficiency criterion was not satisfied for all XAI methods, but also did not change for the different model configurations. Therefore, it is assumed that Differential Privacy and Federated Learning do not impact whether these criteria are satisfied or not. The results for the Faithfulness Estimate metric are displayed in Figure 4.3. Here, the scale was adjusted to view only the most relevant section of the plot that does not show some outliers. The full plot with all outliers and the significance plots for both the logistic and the neural model can be view in Appendix A.2.

Furthermore, no general trend can be observed with this metric. This also shows in the statistical significance, since almost no differences between the Faithfulness evaluations of the different model configurations were significant. Therefore, neither Differential Privacy nor Federated Learning have an effect on the quality of the explanations regarding their Faithfulness according to the results presented.

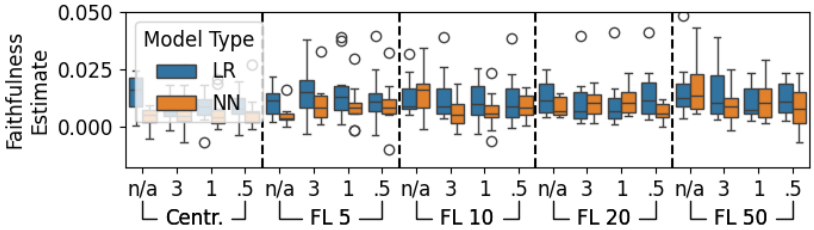


Figure 4.3: Faithfulness Estimate for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

4.2.2 Robustness

For evaluating the explanations on their Robustness, the XAI methods' attributions were qualified with the Local Lipschitz Estimate, average and maximum Sensitivity metrics. The results of the evaluations are visualised in Figure 4.4 alongside with their statistical significance in Figure 4.5. Again, for Figure 4.4 the scale for the Sensitivity metrics is adjusted to highlight the differences of the relevant sections and the full plot with all outliers can be viewed in Appendix A.2. Furthermore, only the significance for the maximum Sensitivity for the logistic model and the Local Lipschitz Estimate for the neural model are displayed, since no general significant trend could be observed for the other metrics. All omitted visualisations ca be inspected in Appendix A.2.

The Robustness metrics exhibit similar results for each model configuration of the logistic and neural model. However, there are some interesting findings that need to be pointed out as the statistical significance of the difference between the results is explored. For one, there are differences for the maximum Sensitivity in the centralised logistic model between the baseline with no application of Differential Privacy and the models utilising a privacy budget. Here, using Differential Privacy increases the maximum Sensitivity significantly compared to not setting a privacy budget. Hence, the Robustness of explanations of the centralised model are affected and are not as robust as if no privacy budget is set. Nevertheless, this phenomenon is not transferable to the application of Federated Learning. When Differential Privacy and Federated Learning are combined, the difference of the maximum Sensitivity is not significant for all privacy budgets when compared to not utilising Differential Privacy in that setting. However, this is not observable for the neural model. Hence, Differential privacy can make explanations less robust by increasing the maximum Sensitivity, but Federated Learning can mitigate these negative effects. The ability of Federated Learning mitigating negative effects of Differential Privacy is also observed in the literature [9].

Furthermore, Federated Learning in combination with Differential Privacy is even lowering the maximum Sensitivity of the logistic model above a certain threshold of clients. When comparing

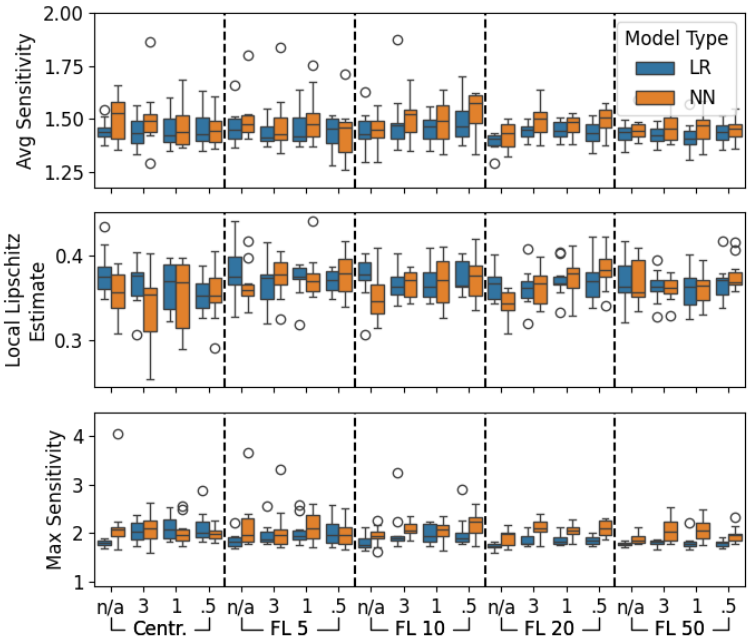
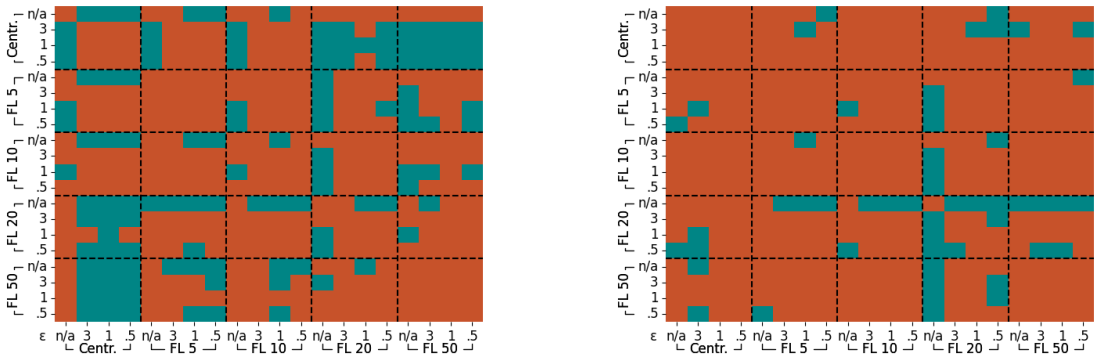


Figure 4.4: Robustness metrics for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



(a) LR Max Sensitivity

(b) NN Local Lipschitz Estimate

Figure 4.5: Student's paired t-test of the Robustness metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

the centralised logistic models that utilised a privacy budget with the same models optimised with Federated Learning with 20 and 50 clients, one can observe that all models are evaluated with lower maximum Sensitivity compared to their centralised counterparts. Consequently, this highlights the possibility of Federated Learning resolving negative effects of Differential Privacy on explainability.

Interestingly, the average Sensitivity is not affected by the described phenomenon, hinting that Federated Learning can only mitigate negative effects of Differential Privacy in a few cases. Hence, further emphasising that this mitigation is not generally applicable, but still can be observed sparsely.

Overall, the Robustness of explanations are mostly unaffected by utilising Differential Privacy or Federated Learning, but some negative effects of Differential Privacy can be mitigated by combining it with a Federated Learning approach.

4.2.3 Complexity

The Complexity of the explanations given by XAI methods are evaluated with the Effective Complexity, Complexity and Sparseness metrics. As for the Sparseness of the explanations, for LIME most values are between 0.99 and 1.0 except for one outlier below .8. Therefore, no general trend could be identified and the visualisation of the Sparseness results are omitted. Results of the Effective Complexity and Complexity evaluations are visualised in Figure 4.6 with Figure 4.7 providing context by showing the pairwise statistical significance.

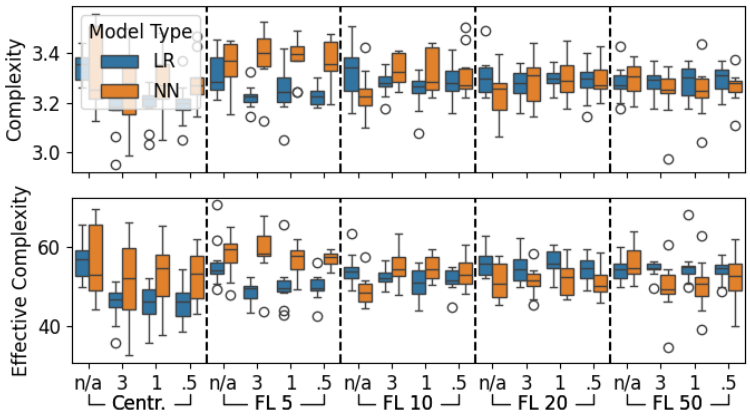
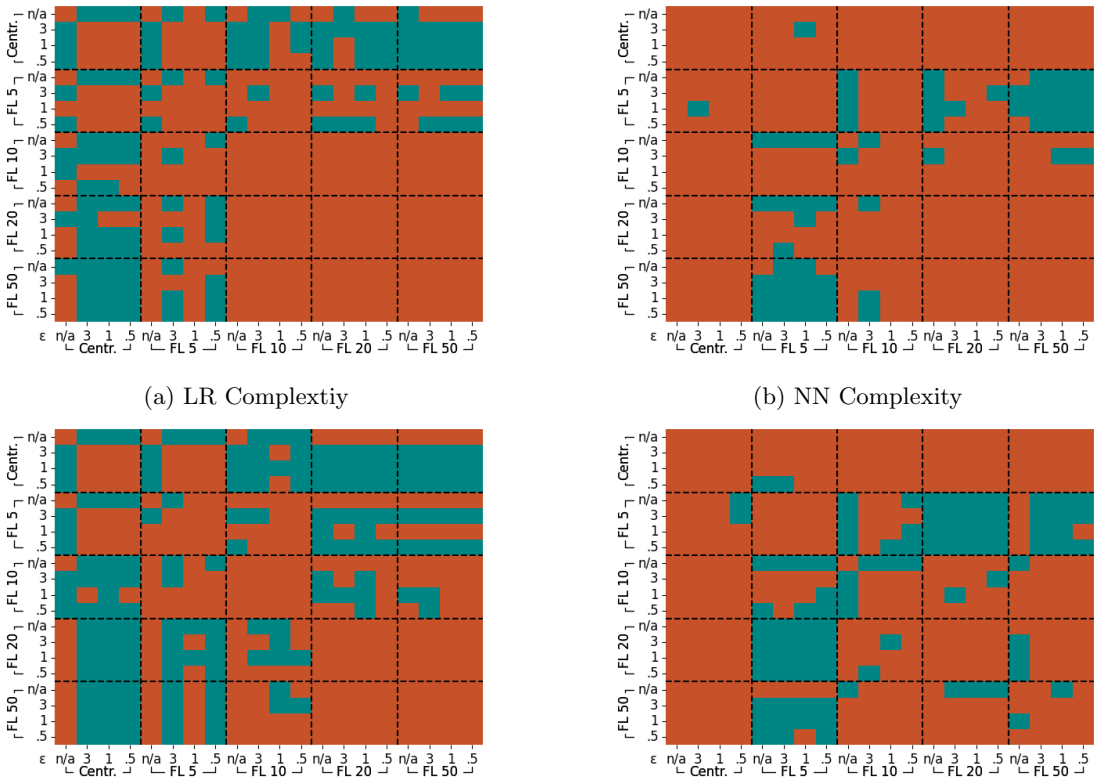


Figure 4.6: Complexity metrics for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

For the centralised logistic model, the general application of Differential Privacy resulted in significantly reducing both the Complexity and Effective Complexity compared to the baseline not utilising Differential Privacy. However, this effect cannot be observed once Federated Learning is combined with Differential Privacy. Furthermore, changes in the Complexity are only apparent for the logistic model and not the neural model. Consequently, applying Differential Privacy on its own can results in less complex explanations for some cases, but it does not have a reducing effect on Complexity in general.

In contrast, Federated Learning paired with Differential Privacy even increases the Complexity and Effective Complexity significantly for most logistic models when compared to the counterparts of the centralised model. Comparing the centralised models with a privacy budget to the federated models with only 5 clients, no significant differences are apparent. However, when more clients are used for the distributed training, both Complexity and Effective Complexity increase with more significant differences compared to the centralised model. Hence, using more clients when combining Differential Privacy with Federated Learning can increase the overall Complexity of explanations. Still, this difference is only observable with the logistic model and not the neural model and therefore cannot be generalised. Nevertheless, this hints to undesirable effects resulting from the combination of Differential Privacy and Federated Learning.



(a) LR Complextiy

(b) NN Complexity

(c) LR Effective Complexity

(d) NN Effective Complexity

Figure 4.7: Student's paired t-test of the Complexity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

Furthermore, Federated Learning without the utilisation of Differential Privacy can even lower the Complexity and Effective Complexity, when comparing the collaborative neural models. Here, both explanations for the federated models with 5 and 50 clients are each significantly more

complex than the explanations for the model optimised with 10 or 20 clients. Therefore, using too few or too many clients when applying Federated Learning can make the explanations for the model more complex. Nonetheless, this difference is only significant for the neural model but not for the logistic one.

Summarising the results of the Complexity of the explanations, no general trend or effect of Differential Privacy, Federated Learning or their combination can be observed as significant. However, for some cases, Differential Privacy and Federated Learning on their own are able to reduce the Complexity for different model types. Furthermore, combining these two techniques can even result in undesirable effects by increasing the overall Complexity of explanations in some cases. Overall, the effects observed on the Complexity of the explanations are dependent on the model type.

4.2.4 Sensitivity

Explanations given by the XAI methods are evaluated on their Sensitivity with MPRT, Efficient MPRT and Smooth MPRT. The evaluation results are depicted in Figure 4.8 contextualised with the pairwise statistical significance in Figure 4.9. Overall, the results do not show a clear general trend for neither Differential Privacy nor Federated Learning. Especially the pairwise statistical significance for the MPRT and Efficient MPRT metric evaluation results for the neural model did not show a lot of significant results and can be viewed in Appendix A.2.

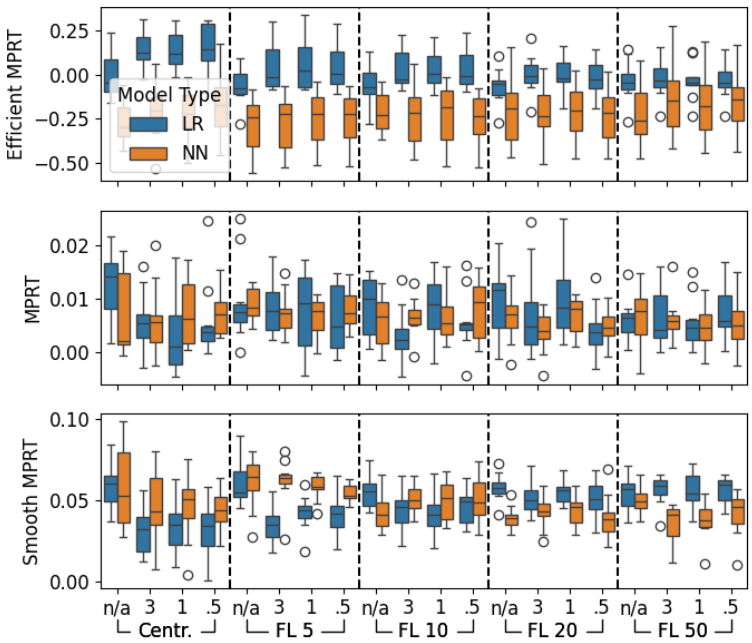
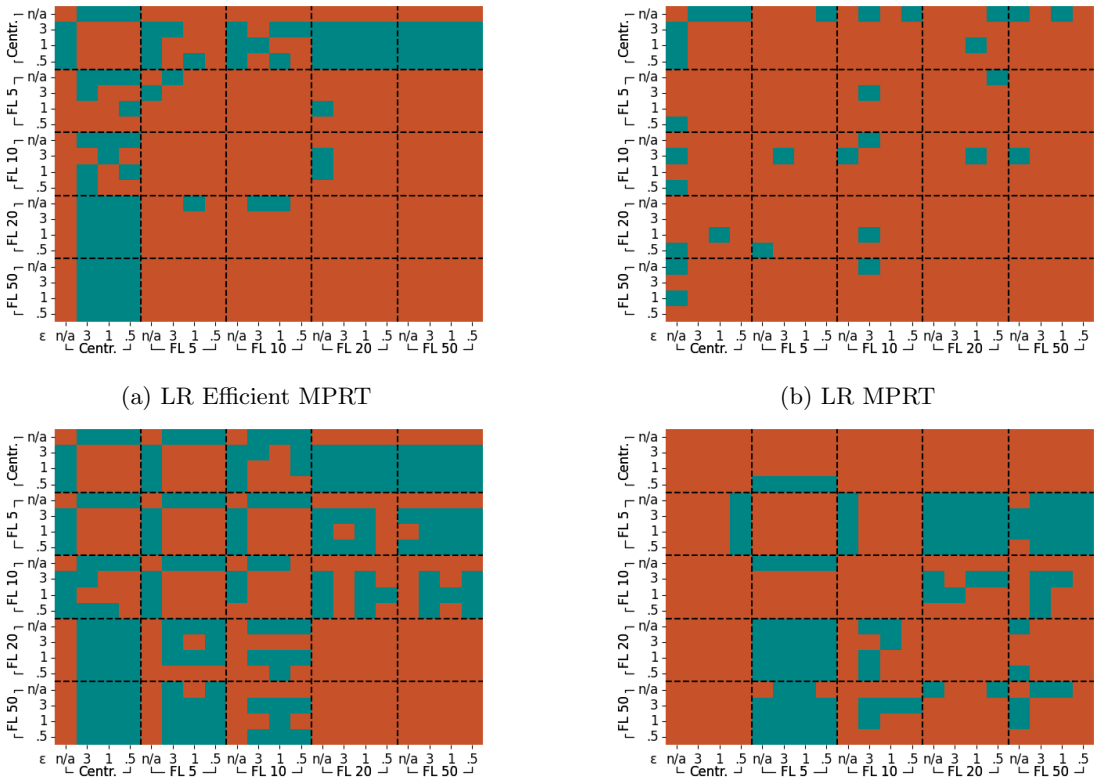


Figure 4.8: Sensitivity metrics for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

For all Sensitivity metrics, the Sensitivity shifted significantly for the centralised logistic model

when comparing the baseline to the models utilising Differential Privacy. Interestingly, when using a privacy budget the Efficient MPRT evaluation scores increased compared to the baseline, whereas the MPRT and Smooth MPRT evaluations scores decreased. However, since the Efficient MPRT metric measures the relative increase in entropy before and after the randomisation of the model parameters and MPRT and Smooth MPRT measure the correlation between the original explanations and the explanations given for the altered model, all metrics hint towards the same phenomenon of explanations being more sensitive to changes. Therefore, applying Differential Privacy to the centralised logistic model lowers the overall Sensitivity of the XAI method towards alterations in the model. In contrast, this effect is not observable when exploring the Sensitivity of the explanations for the neural model and therefore hinting to a situational phenomenon.



(a) LR Efficient MPRT

(b) LR MPRT

(c) LR Smooth MPRT

(d) NN Smooth MPRT

Figure 4.9: Student's paired t-test of the Sensitivity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

For the Smooth MPRT evaluation results, this effect of lowering the Sensitivity furthermore translates to the Federated Learning models with 5 and 10 clients. Here, the Sensitivity is decreasing significantly mostly as well when utilising Differential Privacy in comparison to the

respective federated models not using a privacy budget. Hence, Differential Privacy is able to reduce the Sensitivity for the logistic model not only in the centralised case, but also for some federated models. Still, using different privacy budgets are not affecting the Sensitivity significantly.

In contrast, a significant increase in Sensitivity of the explanations can be observed, when the Federated Learning models utilising Differential Privacy were optimised with 20 or more clients. Here, a significant increase in the Smooth MPRT and decrease in the Efficient MPRT evaluation results are apparent, therefore hinting towards lower entropy of the explanations and higher correlation between the explanations of the original and the altered model. However, the opposite effect is evident when comparing the federated neural models with 5 clients to the ones with 20 and 50 clients. Here, opposite to the logistic model, a significant decrease in the Smooth MPRT and increase in the Efficient MPRT evaluation results are measured. Therefore, depending on the model and the number of clients, Federated Learning can have both negative and positive effects on the Sensitivity of explanations in combination with Differential Privacy.

In summary, the effect of Differential Privacy and Federated Learning on the Sensitivity of the explanations cannot be generalised. Nevertheless, situational phenomenons can be observed where Differential Privacy is able to reduce Sensitivity depending on the model type. Furthermore, Federated Learning can both significantly increase and decrease the Sensitivity for different models.

4.2.5 Axiomatic

Two axiomatic metrics are utilised to qualify the explainability of the different models, namely Completeness and Non-Sensitivity. For all XAI methods, the Completeness axiom was not satisfied and did not change with neither Federated Learning nor Differential Privacy. Therefore, it is assumed that these privacy-preserving techniques do not have an impact on the Completeness of explanations. When evaluating LIME for Non-Sensitivity, no general trend could be discovered and the differences between the different parameters for Federated Learning and Differential Privacy also did not have a significant impact for almost all the models. Hence, also Non-Sensitivity is apparently not impacted by Differential Privacy or Federated Learning.

4.3 Privacy Evaluation of Differential Privacy and Federated Learning

To explore the privacy aspect of the privacy-explainability trade-off, the Accuracy, Precision and Recall of the MIA with the shadow model technique as described in Section 2.2.2 resulting from the experiments are visualised in Figure 4.10. These results are contextualised with the pairwise statistical significance in Figure 4.11. Since the Recall did not show any significant differences between the pairwise comparison, the visualisation of the pairwise statistical significance of the differences of the Recall of the attacks is omitted and can be viewed in Appendix A.3.

The Accuracy and Recall evaluation results show a similar picture of a seemingly horizontal line at 0.5 hinting that in general an attack in the setting of the experiment is challenging. Furthermore, as the Accuracy for both the logistic and neural model in Figure 4.10 show, there is no recognisable overarching trend that shows an impact of either Differential Privacy or Federated Learning on the success of the attack.

Additionally, there is no significant recognisable trend in the Recall evaluation results and therefore neither Federated Learning nor the application of Differential Privacy have an effect on the ability of the attack to extract samples used in the original training dataset.

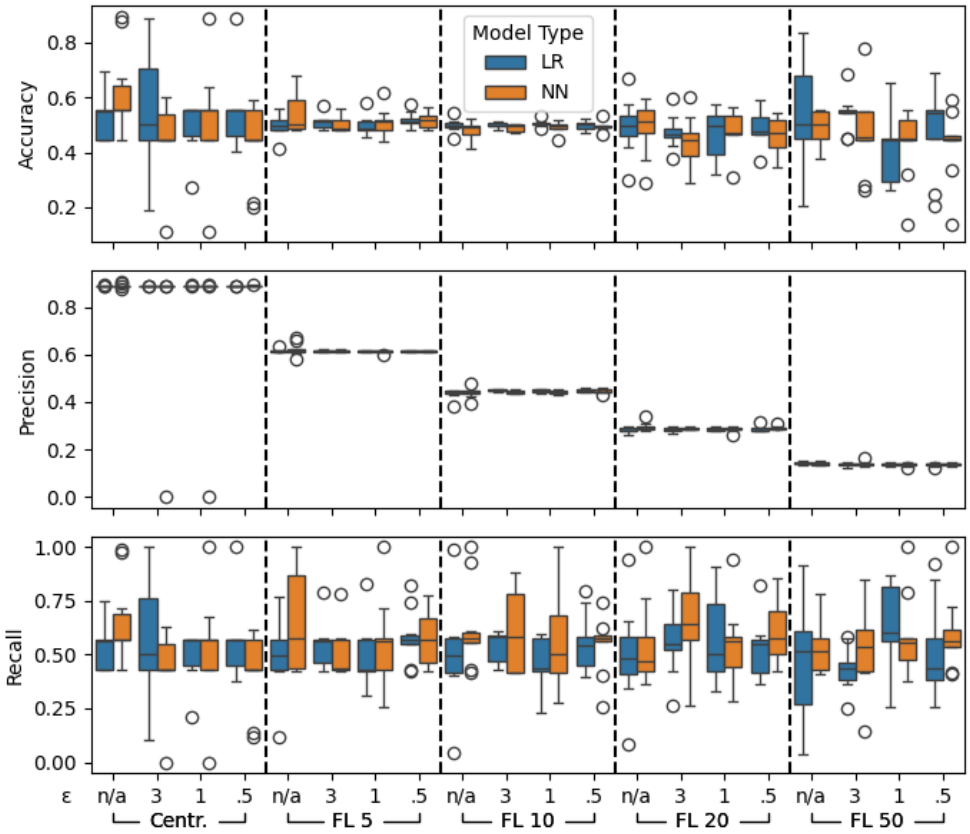
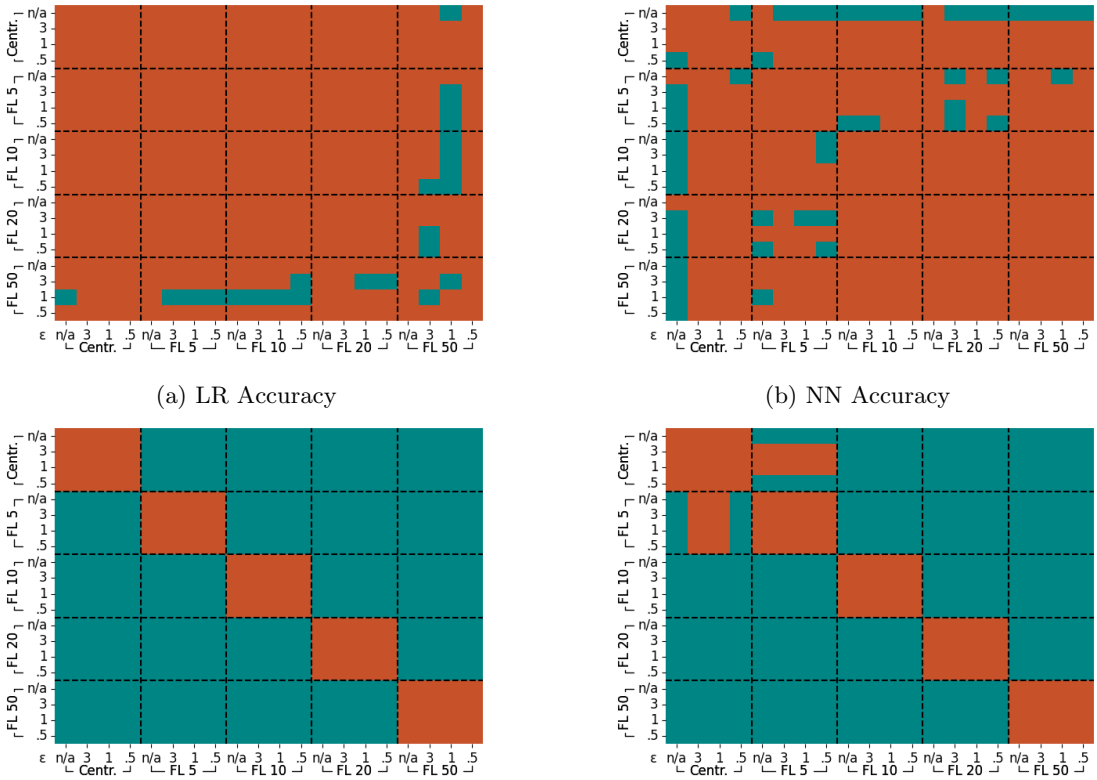


Figure 4.10: MIA success rate (Accuracy, Precision, Recall) of each trained model grouped by their model type. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

However, for the Precision of the attack, a clear trend can be observed for the federated models. Here, using more clients has a strong and significant impact on the Precision, lowering the scores by more than 0.1 from each group to the next with a higher number of clients. Ultimately, the Precision is lowered from over 0.8 for the centralised model to less than 0.2 for Federated Learning with 50 clients. This effect is not significant for all models utilising Differential Privacy that are optimised with 5 clients compared to their centralised counterparts, showing one exception to this rule. Nevertheless, Federated Learning is showing a strong effect in its ability to protect the training data from extraction by an attacker using the shadow model technique.

Overall, Differential Privacy does not have an impact on the success of the MIA utilised in this work. However, Federated Learning is able to lower the Precision significantly by using more clients during the training process.



(a) LR Accuracy

(b) NN Accuracy

(c) LR Precision

(d) NN Precision

Figure 4.11: Student's paired t-test of each MIA metric for comparison with counterparts. Green indicates significant ($p < 0.05$) results; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

4.4 Summary

In summary, the impact of Differential Privacy on neural models is more apparent compared to logistic models, highlighting a dependency on model type. The use of Differential Privacy can either improve or worsen model performance, depending on the specific metrics and model types. Federated models show negligible performance changes with an increasing number of clients until a certain threshold, where for the logistic model the performance decreases.

No significant differences are found in the Faithfulness evaluations across different model configurations, indicating that Differential Privacy and Federated Learning do not affect the quality of explanations regarding their Faithfulness.

The Robustness of explanations remains largely unaffected by these techniques, though some negative effects of Differential Privacy can be alleviated when combined with Federated Learning.

Regarding the Complexity of explanations, no consistent trend or significant effect of Dif-

ferential Privacy, Federated Learning, or their combination is observed. In some cases, these techniques individually reduce Complexity for certain models, but their combination can sometimes increase it. The effects on Complexity are thus model-dependent.

The Sensitivity of explanations varies in some cases with the use of Differential Privacy and Federated Learning, depending on the model type. Differential Privacy can reduce Sensitivity in some scenarios, while Federated Learning can either increase or decrease Sensitivity significantly for different models. However, no general trends could be observed.

The Completeness axiom was not satisfied by any XAI method, and neither Differential Privacy nor Federated Learning influenced this outcome. Similarly, the Non-Sensitivity evaluation showed no general trend or significant impact of these privacy-preserving techniques across most models.

Finally, Differential Privacy does not affect the success of the MIA used in this study, whereas Federated Learning significantly reduces the Precision of MIA with an increasing number of clients.

The next chapter will give a detailed discussion of these findings, providing a deeper analysis and interpretation of the results.

Chapter 5

Discussion and Limitations

In this chapter, the results presented in Chapter 4 are discussed and contextualised with the literature in order to answer the research question posted in Section 1.1. Then, failed approaches are described and the limitations of this work are presented.

5.1 Discussion

This section elaborates on the findings shown in Chapter 4 to answer the research questions stated in Section 1.1. For easier readability, the research questions are restated here:

1. What is the effect of applying Differential Privacy (DP) and/or Federated Learning (FL) on the explainability of a classification model?
2. To what extent can the explainability of models trained with DP and/or FL be preserved, while concurrently minimising the success rate of external attackers attempting to infer information about the training data through a black-box membership inference attack?

5.1.1 Prediction Performance

The impact of Differential Privacy on model performance is notably model-dependent, with neural models being more significantly affected than logistic models. This suggests that the Complexity of the model plays a crucial role in how Differential Privacy influences performance. In particular, the application of Differential Privacy can either improve or worsen performance based on specific metrics and model types. This aligns with previous research indicating that Differential Privacy can reduce performance due to added noise, with the intention of preserving privacy but can hinder model accuracy [8].

Conversely, Federated models exhibit negligible performance changes with an increasing number of clients beyond a certain threshold, indicating that the benefits of FL in terms of distributed data handling and privacy preservation do not necessarily translate to performance loss beyond a certain point.

5.1.2 Explainability

The explainability of models, as assessed through various metrics, shows no general trend when Differential Privacy and Federated Learning are applied. The Faithfulness and Robustness of

explanations remain largely unaffected, suggesting that these privacy-preserving techniques can be tuned without significantly compromising the quality of explanations.

However, some negative effects of Differential Privacy on the Robustness of explanations can be mitigated by combining it with Federated Learning, supporting the notion that Federated Learning can help alleviate certain drawbacks of applying Differential Privacy [9].

The Complexity of explanations also varies depending on the model, with no consistent effect observed from the application of Differential Privacy or Federated Learning. This implies that while these techniques can individually reduce Complexity for some models, their combined application may sometimes increase it. Therefore, making the effects on Complexity highly model-dependent and not generalisable.

The Sensitivity of explanations, also shows not much variability with the use of Differential Privacy and Federated Learning. In some scenarios, Differential Privacy can reduce Sensitivity, whereas Federated Learning can either increase or decrease Sensitivity significantly, depending on the model. These findings underscore the necessity for model-specific tuning, but highlight the applicability of these techniques while maintaining explainability. These findings are in contrast to the results of Saifullah et al. [9], as they found that explainability is reduced when utilising Differential Privacy.

5.1.3 Privacy

In terms of MIAs, the application of Differential Privacy does not affect the success of the attacks, as measured by recall and accuracy. However, Federated Learning significantly reduces the precision of MIA with an increasing number of clients, indicating a higher false positive rate. This suggests that while Federated Learning makes it more difficult to definitively determine whether specific data was part of the training set by reducing Precision, it does not compromise the overall detection capability of the attack through maintaining recall and accuracy. Therefore, it is implied that Federated Learning can make it harder for attackers to be certain about the data they extract, enhancing the privacy aspect without substantially affecting the accuracy of privacy-preserving measures.

5.1.4 Privacy-explainability trade-off

With the results and implications presented, a trade-off between the privacy and explainability by utilising Differential Privacy and Federated Learning cannot be identified. Since the overall effect of these techniques on explainability is evaluated as model-dependant and not generalisable, a trade-off is not apparent. Still, there are effects to be considered when applying Differential Privacy and Federated Learning as previously shown. Moreover, Federated Learning makes an effective tool in preserving privacy by reducing the certainty of an attacker to determine if samples are part of the original training set or not.

5.1.5 Failed approaches

Before using the MIMIC-Extract dataset [67], it was explored if the length of stay prediction yield good results by using the ICD[1] codes and the patients' vital measurements associated with each stay. Furthermore, the ICD codes were encoded into embeddings [71] to provide a bigger and more meaningful feature space before applying 1D pooling to reduce the dimension [72] and feed into a deep neural network. However, this attempt was not successful and therefore the MIMIC-Extract dataset with a logistic and neural model are used.

[1] International Classification of Diseases

Additionally, the embedding representation of the ICD codes pose a different problem for generating meaningful and humanly understandable explanations. Hence, a decoder approach was investigated in order to label the embedding representations with the original ICD codes. This decoder could then be used to decode the explanations resulting from the XAI methods to the ICD codes and therefore giving their relevance to the prediction. This approach only worked, when the ICD codes were reduced to the first 4 letter of the codes and was therefore discarded.

5.2 Limitations

This work has the following limitations. The evaluation was conducted on one specific dataset in the health domain and therefore might not be fully generalisable to other datasets and domains. The quality and availability of this data can impact the predictors performance and therefore alter its explainability and ability to keep data private.

Furthermore, only three XAI methods were investigated for the explainability of the model. Even though the methods used are widely used within the literature and cover different approaches, there might be XAI methods that are able to effectively explain a predictor optimised with a lower privacy budget. Still, the results presented give an indication towards the effect of Differential Privacy and Federated Learning on explainability.

Additionally, only one attack type was investigated, namely MIA with the shadow model technique. Hence, there might be other attacks or techniques that could more effectively extract information about the training data from the predictor. Consequently, the insights provided for this specific attack may not be fully transferable to other attacks and techniques.

Despite these limitations, this work provides valuable insights into the effect of Federated Learning and Differential Privacy on privacy and explainability. The findings contribute to the growing body of research aiming to develop explainable prediction models that mitigate the susceptibility to attacks revealing private data.

Chapter 6

Conclusions

Conclusions are given in this chapter by summarising this work and its implications. Finally, suggestions are made for future research.

6.1 Summary

This work explores the balance between explainability and privacy in machine learning models using Differential Privacy and Federated Learning. The goal is to determine how these privacy-preserving techniques affect model explainability and data protection, and is structured around the research questions posed in Chapter 1.1.

The approach involves models applying Differential Privacy and Federated Learning, evaluating their explainability through various metrics, and testing privacy via MIAs. The experiments utilize the MIMIC-Extract [67] dataset, which is related to medical data privacy concerns, and employ cross-validation to ensure robust results.

Key findings include that the impact of Differential Privacy on model performance is highly model-dependent, with neural models experiencing more significant effects compared to logistic models. This aligns with the notion that the added noise from Differential Privacy can degrade performance, particularly in more complex models. Federated Learning, on the other hand, show minimal performance changes beyond a certain client threshold, indicating that its primary benefit lies in collaborative training while maintaining privacy rather than performance enhancement.

Regarding explainability, neither Differential Privacy nor Federated Learning have a consistent effect across models. While sometimes negative impacts of Differential Privacy on explanation Robustness are mitigated in some cases by combining it with Federated Learning. The overall effects on Complexity and Sensitivity vary depending on the model used. Overall, the explainability of models remain largely unaffected by these privacy techniques, suggesting that their application does not necessarily compromise interpretability.

In terms of privacy, Differential Privacy does not significantly alter the success rate of MIAs, but Federated Learning reduce the Precision of these attacks, resulting in a higher false positive rate. This indicates that while Federated Learning enhances privacy by making it more difficult for attackers to confirm whether specific data is included in the training set, it does not prove the overall effectiveness of privacy measures.

Overall, this work does not identify a clear trade-off between privacy and explainability, as the effects of Differential Privacy and Federated Learning are model-dependent, sparse and not universally applicable. Practitioners should consider these findings when applying Differential

Privacy and Federated Learning, focusing on specific model characteristics and desired outcomes to balance explainability with data privacy effectively.

6.2 Future work

One of the failed approaches of this work was the decoding of ICD code embeddings as part of the explanation. This technique was not fully explored, but could have potential for more accurate predictions by providing an extended feature space without limiting the explanation capabilities. Therefore, exploring this potential could be interesting to examine.

This work investigates how privacy and explainability are affected by applying Differential Privacy and Federated Learning by using a dataset from the medical domain. Therefore, it would be interesting to extend this work to other domains where privacy and explainability are vital.

Furthermore, the dataset is distributed equally to all clients during Federated Learning. However, this does not reflect the real world, where the data of each client should each have their own distribution according to the original idea of Federated Learning [24]. For future experiments, an investigation on an already naturally distributed dataset could be of value.

In this work, the attack method used is a MIA using the shadow model technique. However, there are more techniques to implement a MIA that could be worth investigating. Additionally, the effect of Differential Privacy and Federated Learning on different attacks, e.g. model extraction attacks, could be interesting to investigate.

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Appendix A

A.1 Explainability evaluation of Integrated Gradients and Feature Permutation

A.1.1 Integrated Gradients

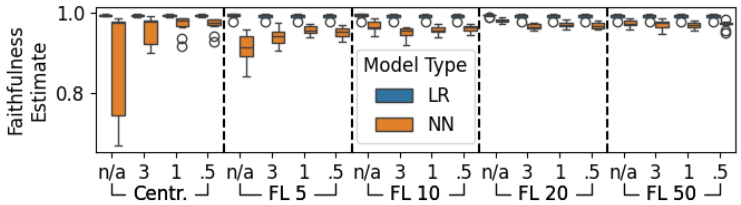
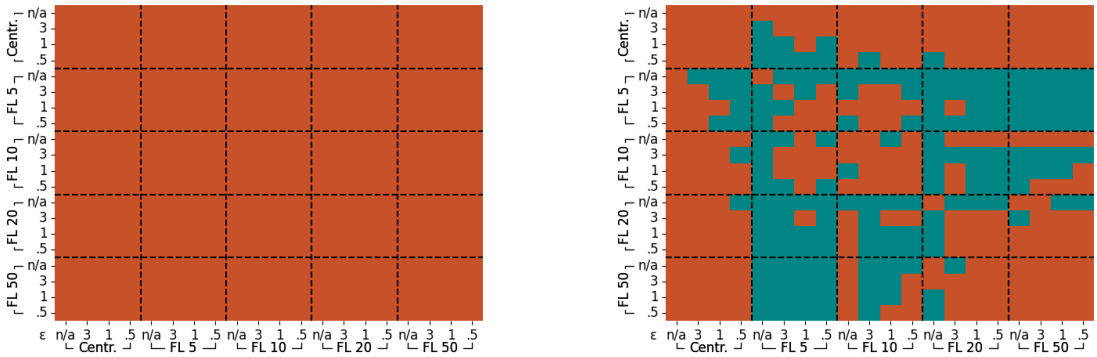


Figure A.1: Faithfulness metrics for models grouped by their model type explained with Integrated Gradients. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



(a) LR Faithfulness Estimate

(b) NN Faithfulness Estimate

Figure A.2: Student's paired t-test of the Faithfulness Estimate of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

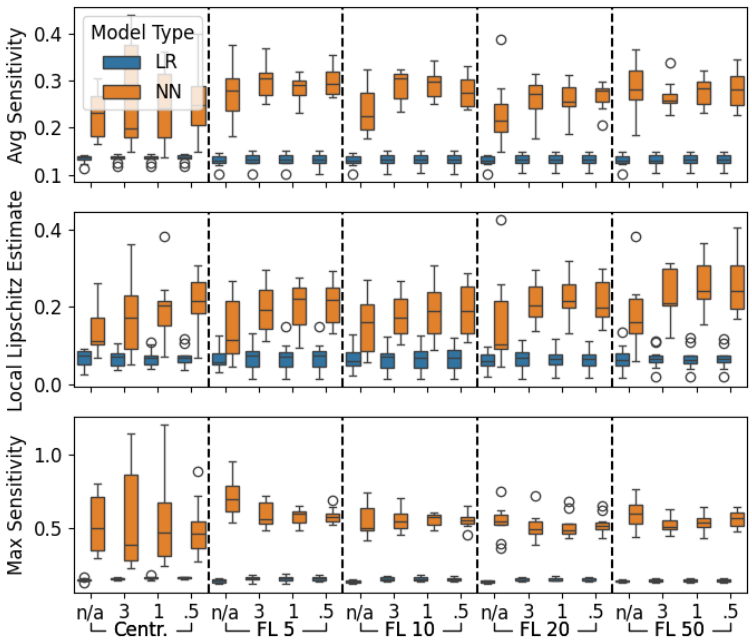
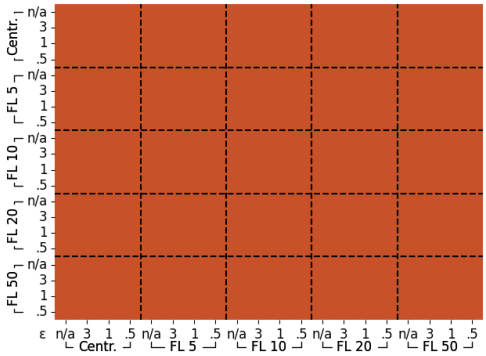
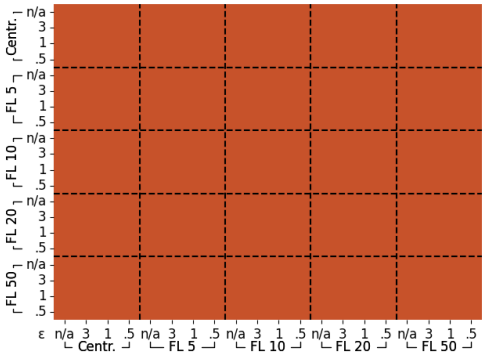


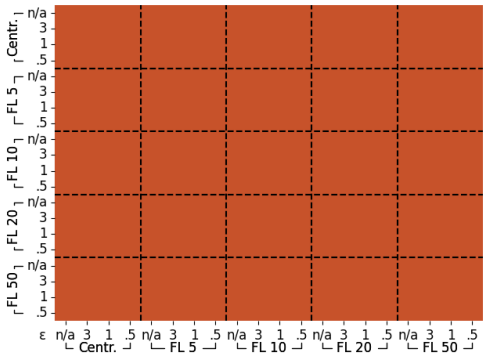
Figure A.3: Robustness metrics for models grouped by their model type explained with Integrated Gradients. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



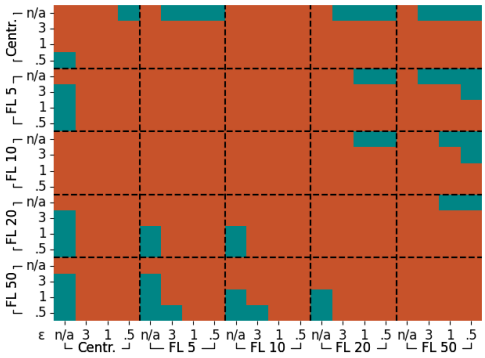
(a) LR Avg Sensitivity



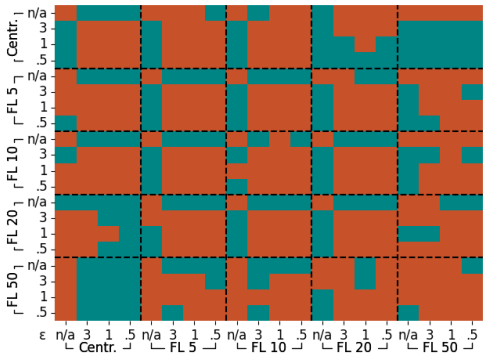
(b) NN Avg Sensitivity



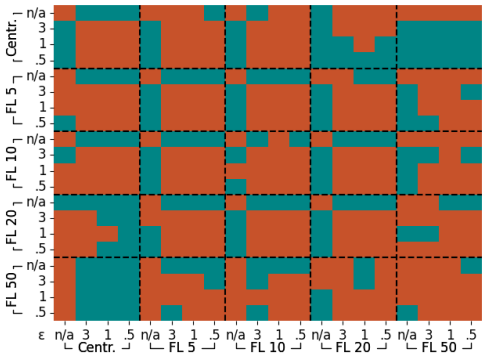
(c) LR Local Lipschitz Estimate



(d) NN Local Lipschitz Estimate



(e) LR Max Sensitivity



(f) NN Max Sensitivity

Figure A.4: Student's paired t-test of the Robustness metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

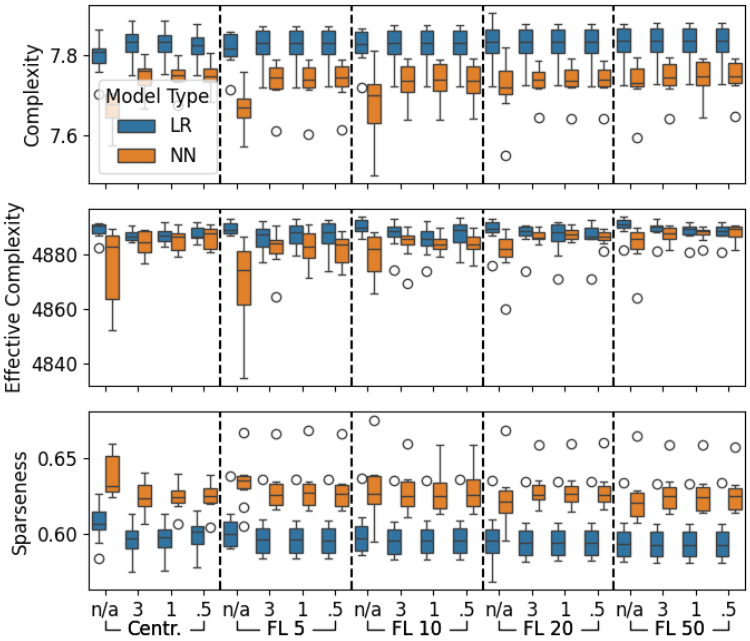
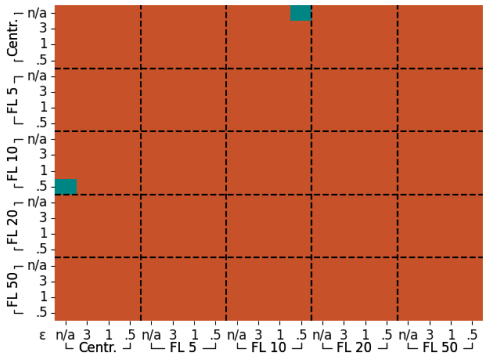
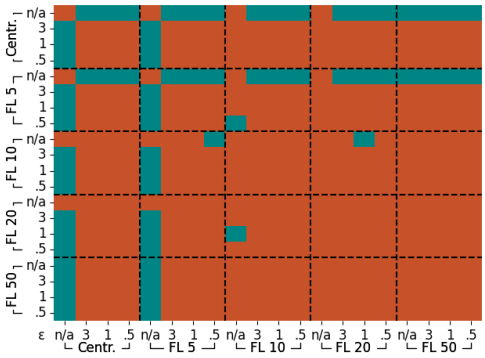


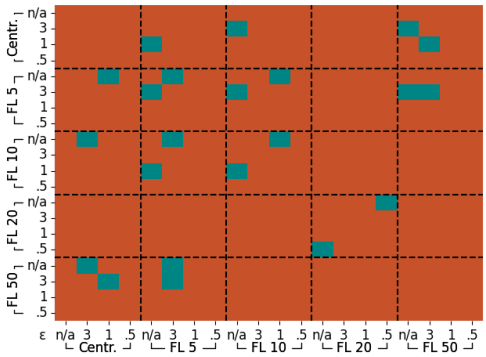
Figure A.5: Complexity metrics for models grouped by their model type explained with Integrated Gradients. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



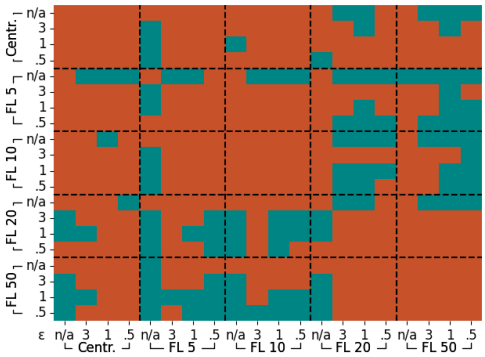
(a) LR Complexity



(b) NN Complexity



(c) LR Effective Complexity



(d) NN Effective Complexity

Figure A.6: Student's paired t-test of the Complexity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

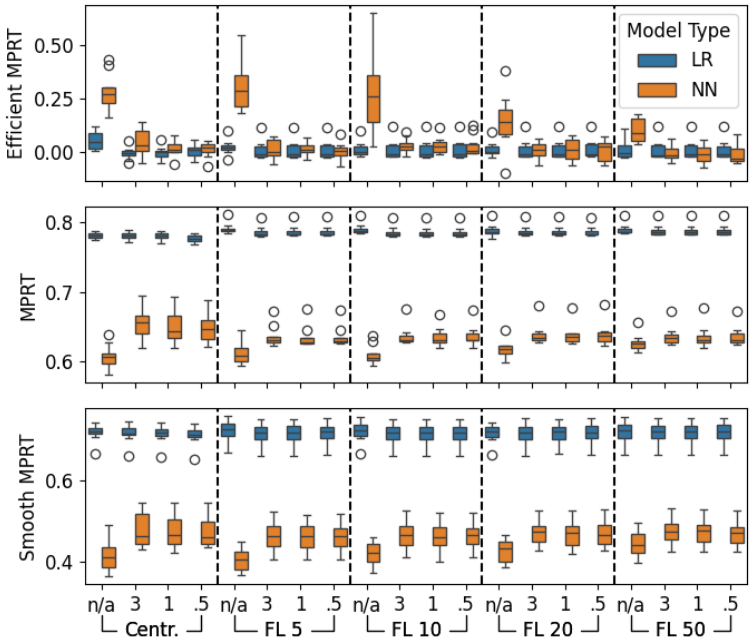
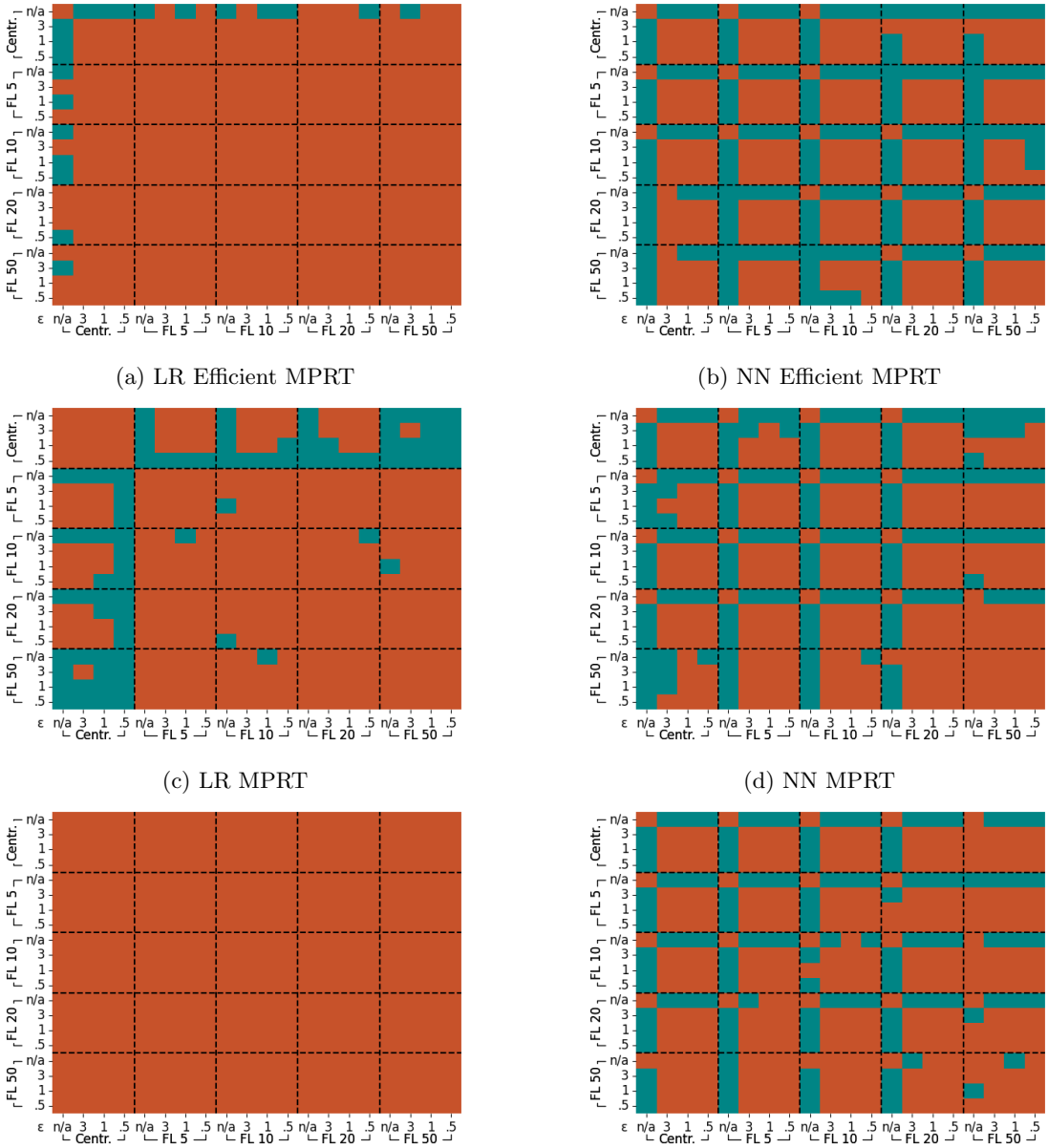


Figure A.7: Sensitivity metrics for models grouped by their model type explained with Integrated Gradients. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



(a) LR Efficient MPRT

(b) NN Efficient MPRT

(c) LR MPRT

(d) NN MPRT

(e) LR Smooth MPRT

(f) NN Smooth MPRT

Figure A.8: Student's paired t-test of the Sensitivity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

A.1.2 Feature Permuation

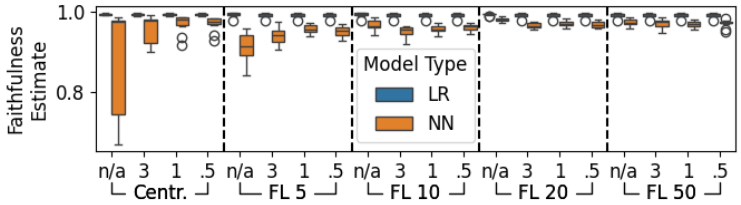
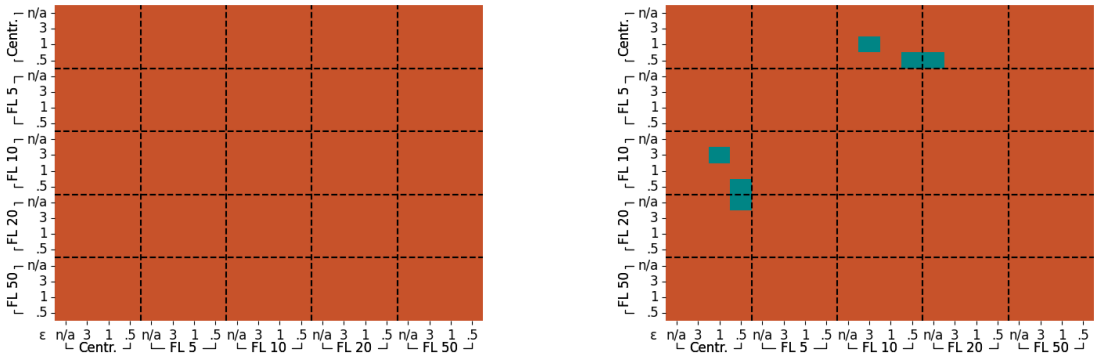


Figure A.9: Faithfulness metrics for models grouped by their model type explained with Feature Permutation. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



(a) LR Faithfulness Estimate

(b) NN Faithfulness Estimate

Figure A.10: Student's paired t-test of the Faithfulness Estimate of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

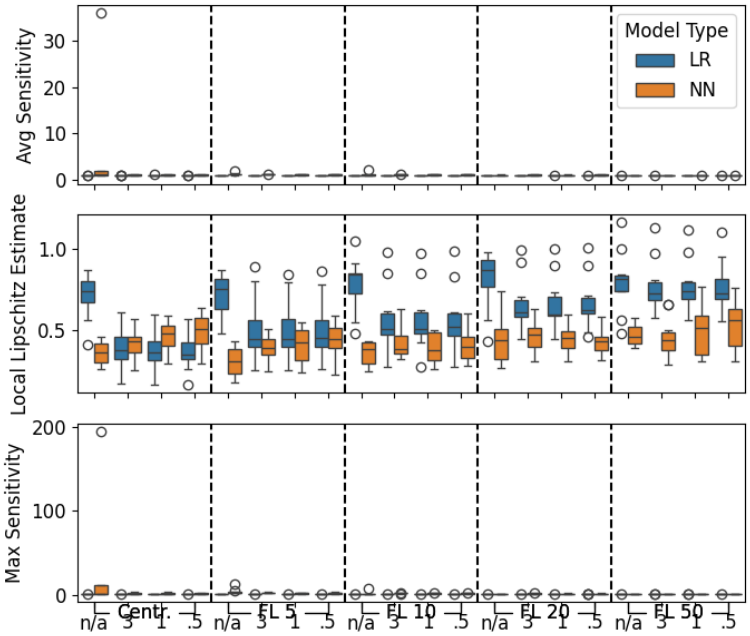
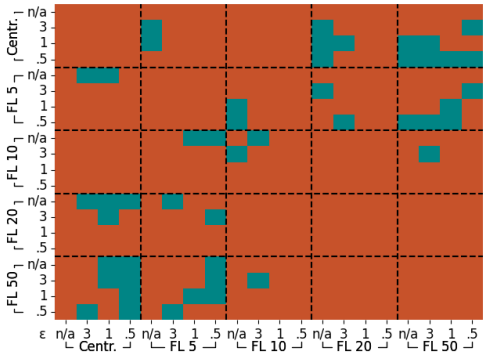
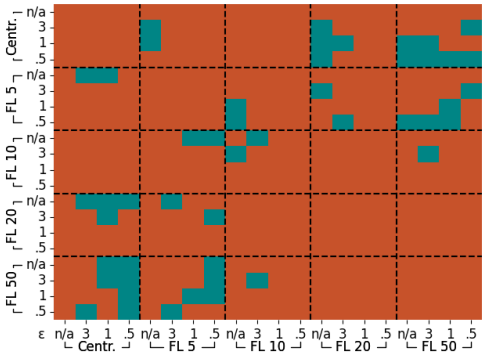


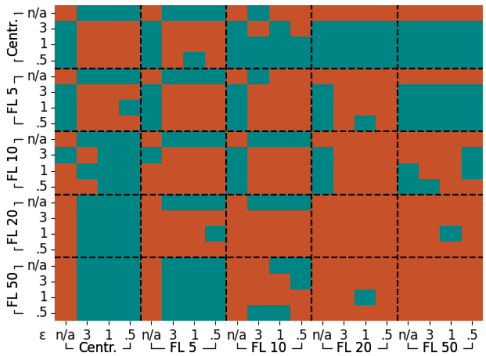
Figure A.11: Robustness metrics for models grouped by their model type explained with Feature Permutation. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from hFPhest to lowest. "n/a" is representative for when Differential Privacy was not used.



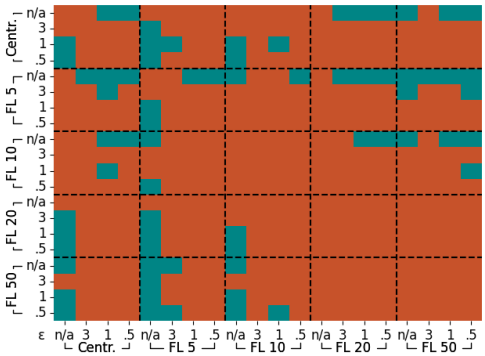
(a) LR Avg Sensitivity



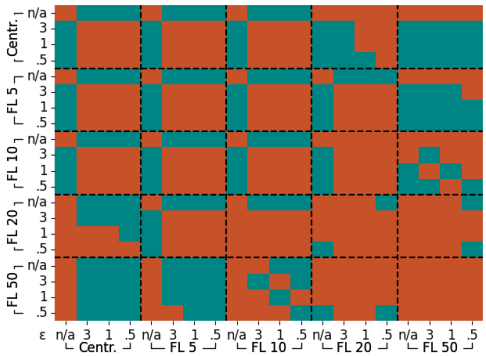
(b) NN Avg Sensitivity



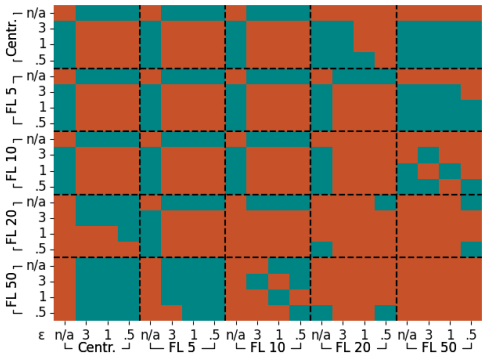
(c) LR Local Lipschitz Estimate



(d) NN Local Lipschitz Estimate



(e) LR Max Sensitivity



(f) NN Max Sensitivity

Figure A.12: Student's paired t-test of the Robustness metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

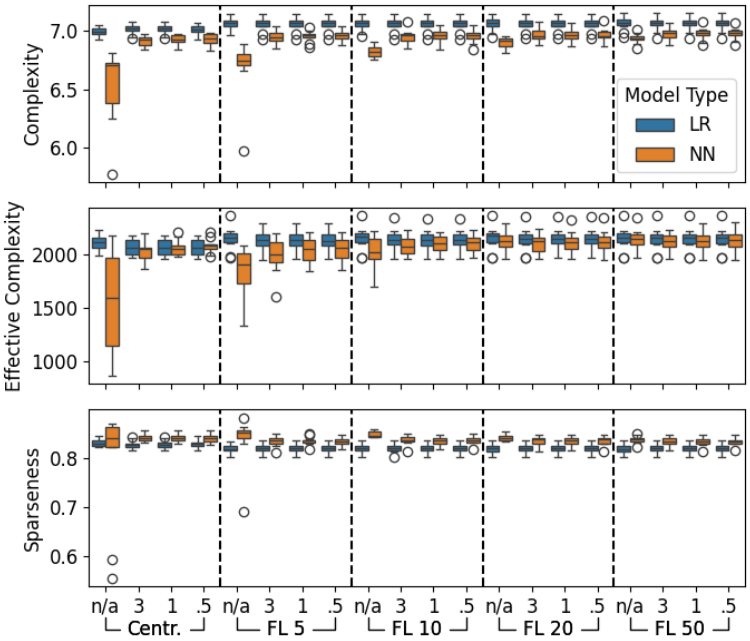
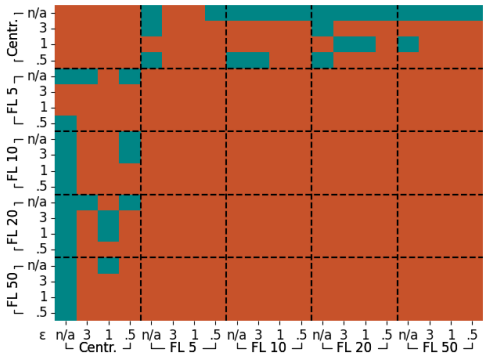
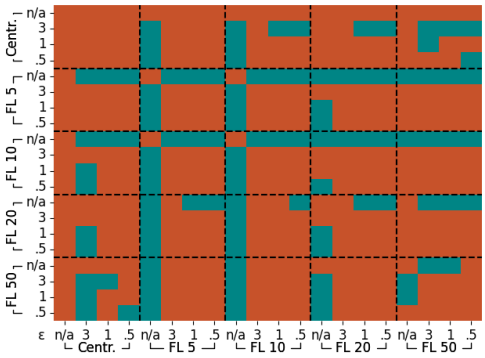


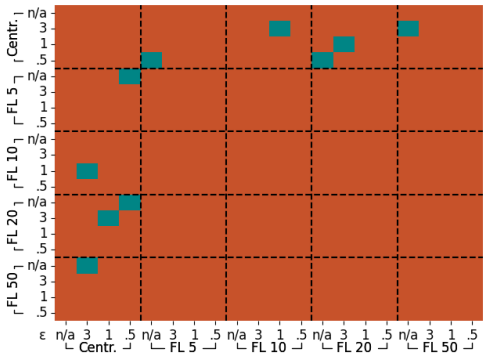
Figure A.13: Complexity metrics for models grouped by their model type explained with Feature Permutation. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



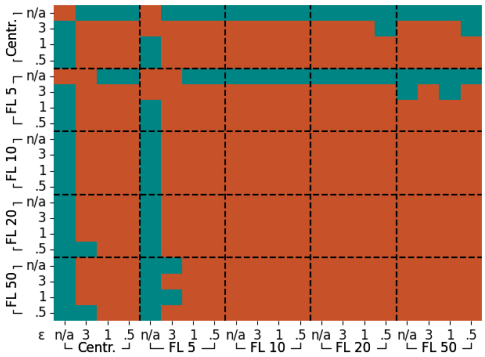
(a) LR Complexity



(b) NN Complexity



(c) LR Effective Complexity



(d) NN Effective Complexity

Figure A.14: Student's paired t-test of the Complexity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

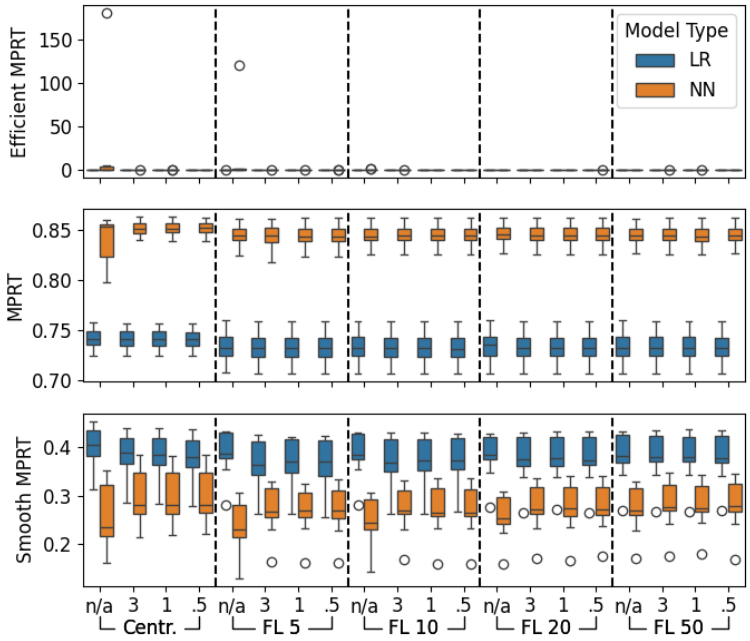
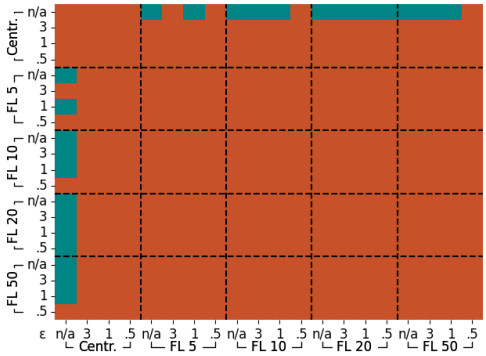
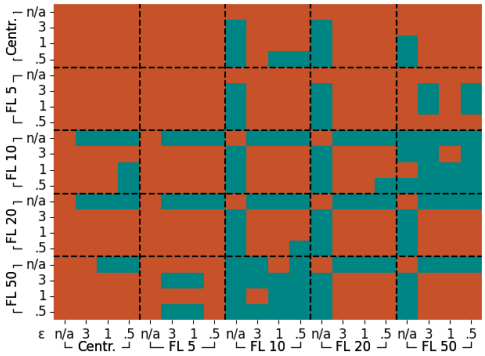


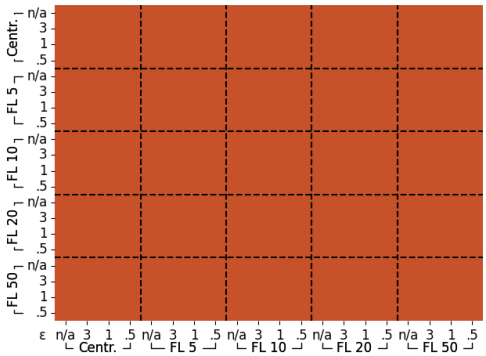
Figure A.15: Sensitivity metrics for models grouped by their model type explained with Feature Permutation. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



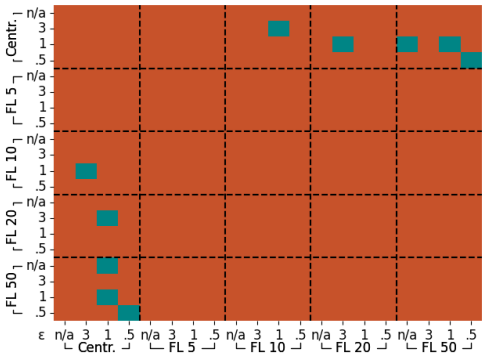
(a) LR Efficient MPRT



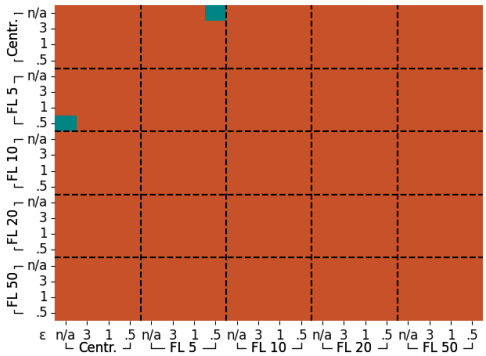
(b) NN Efficient MPRT



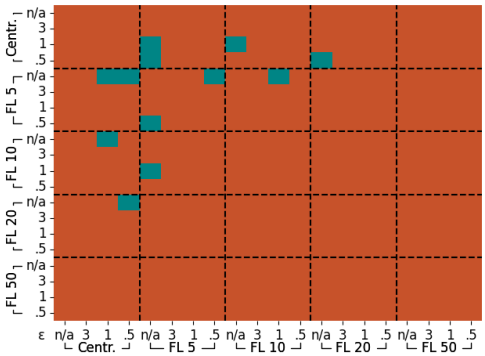
(c) LR MPRT



(d) NN MPRT



(e) LR Smooth MPRT



(f) NN Smooth MPRT

Figure A.16: Student's paired t-test of the Sensitivity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

A.2 Additional results for LIME

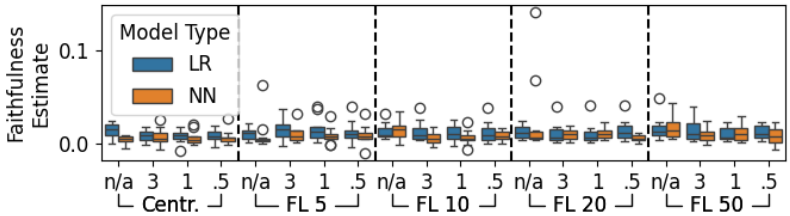


Figure A.17: Faithfulness Estimate with outliers for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.

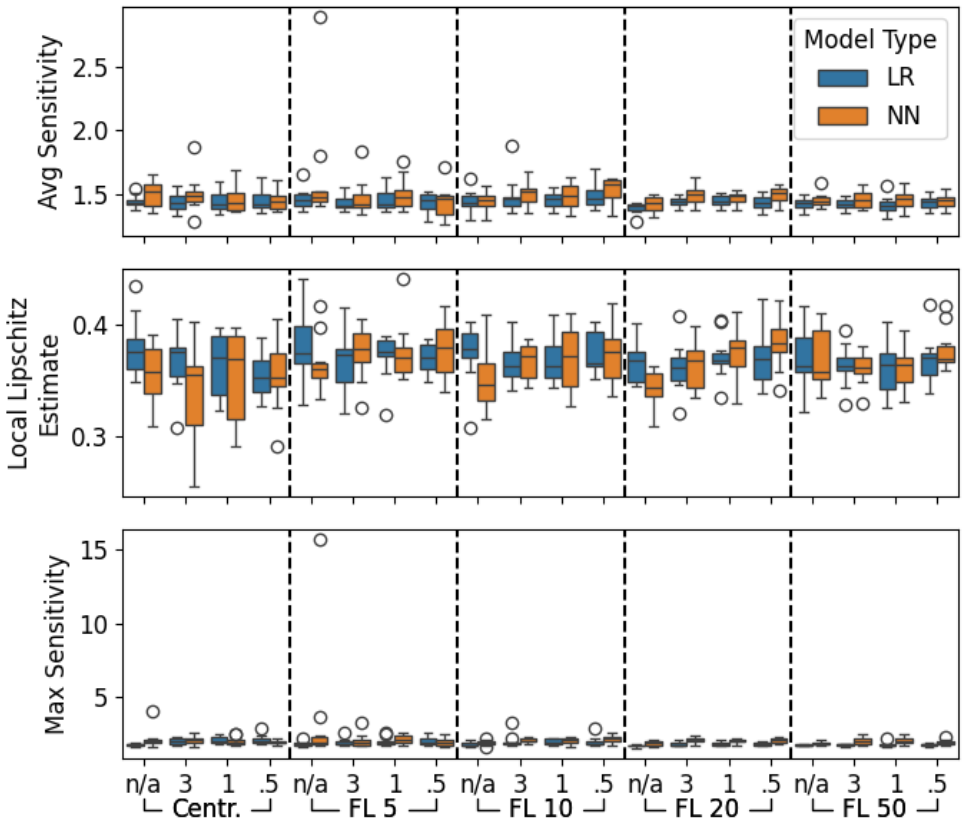
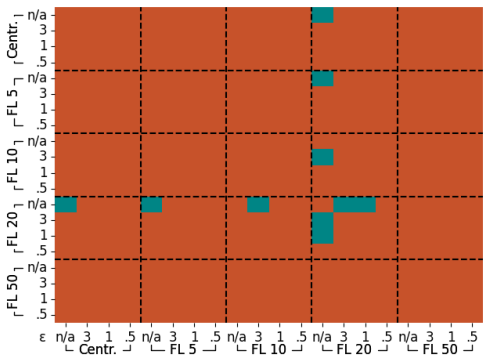
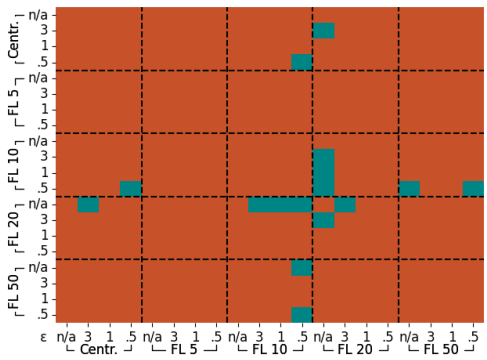


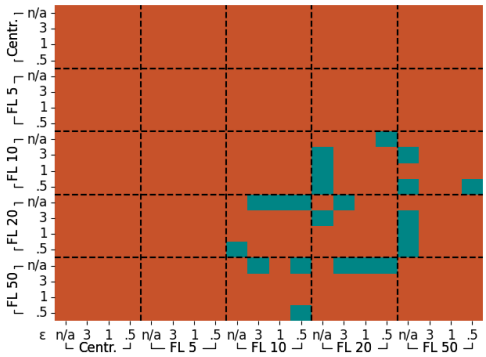
Figure A.18: Robustness metrics with outliers for models grouped by their model type explained with LIME. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.



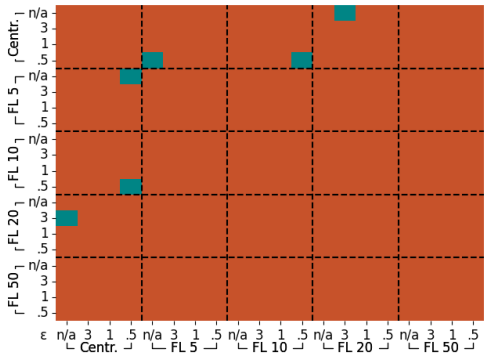
(a) LR Avg Sensitivity



(b) NN Avg Sensitivity

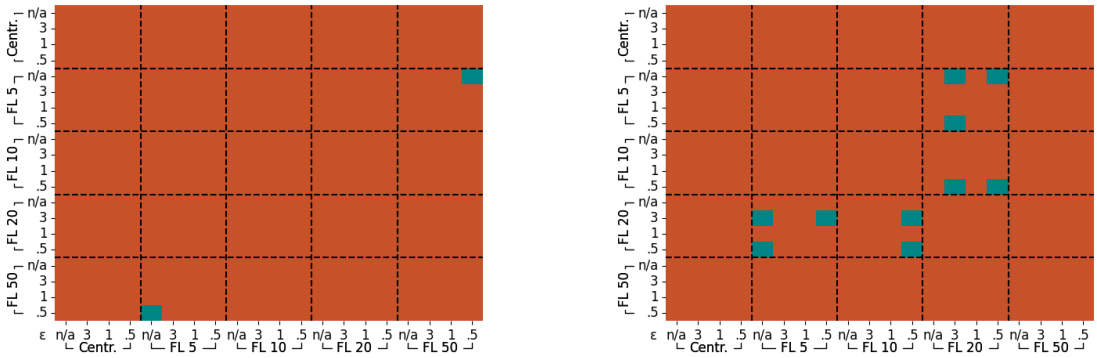


(c) NN Max Sensitivity



(d) LR Local Lipschitz Estimate

Figure A.19: Additional student's paired t-test of the Robustness metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

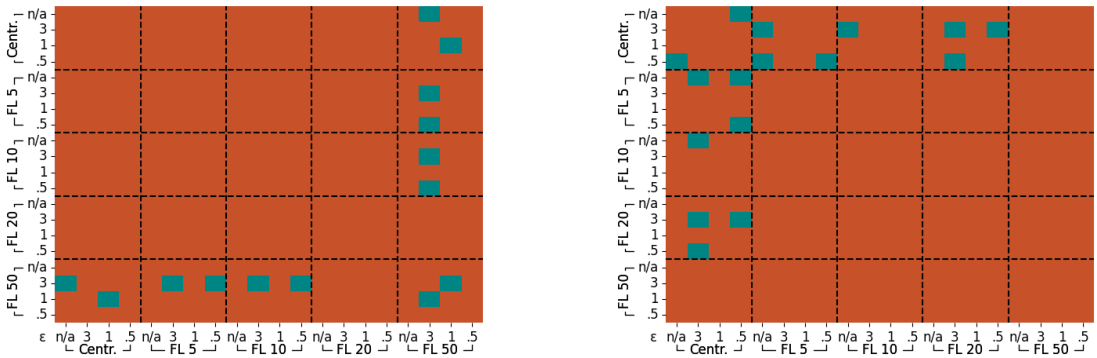


(a) NN Efficient MPRT

(b) NN MPRT

Figure A.20: Additional student's paired t-test of the Sensitivity metrics of each model configuration for comparison with counterparts, where a trend can be recognised. Green indicates significant ($p < 0.05$) differences; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used. Furthermore, the logistic model (LR) is considered separately from the neural model (NN).

A.3 Privacy Evaluation of Differential Privacy and Federated Learning



(a) LR Recall

(b) NN Recall

Figure A.21: Student's paired t-test of the MIA Recall for comparison with counterparts. Green indicates significant ($p < 0.05$) results; red for non-significant. The models are grouped by the number of clients used for Federated Learning (FL), where "Centr." denotes the centralised model. Each group consists of a logistic (LR) and neural (NN) model with 4 different ϵ as privacy budget sorted from highest to lowest. "n/a" is representative for when Differential Privacy was not used.Furthermore, the logistic model (LR) is considered separately from the neural model (NN).